

VERSION 01



LIKAS

USER MANUAL

A COMPREHENSIVE GUIDE FOR USERS

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Welcome Users!

Welcome to Likas, a Flood Rescue Priority Decision Support System (DSS) designed specifically for the local government units and barangays of Manila. Our goal is to streamline the reporting of flood incidents and provide a centralized dashboard for city administrators to effectively prioritize rescue and relief operations based on data-driven insights.

Getting Started

When you use Likas for the first time, there are a few things you're going to need to know before you begin.

Getting Started

This guide tells you how to run the frontend and backend servers yourself using the source code we have provided.

System Requirements

Software Requirements

- Node.js: v18 or the latest stable release.
- npm: The latest version of the Node Package Manager.
- Frontend: React.js.
- Backend Application: Built using Express.js
- Database: PostgreSQL (Supabase)

Hardware Requirements

- A computer or laptop running Windows, Linux, or macOS.
- Minimum 4GB RAM.
- Minimum 10GB free disk space.
- Stable internet connection for map rendering and real-time database syncing.

Interface Requirements

External Hardware Requirements

- A computer or laptop running Windows, Linux, or macOS.
- Minimum 4GB RAM.
- Minimum 10GB free disk space.
- Stable internet connection for map rendering and real-time database syncing.

Installation and Setup

End User Access (Website)

1. Open a web browser and go to **<https://likas.onrender.com/>**
2. Enter your **assigned email address** and **password** on the login screen.
3. Click **Sign In** and wait for the dashboard to fully load before navigating to another page.
4. Once logged in, confirm that the **sidebar menu** and **profile details** match your assigned role (e.g., Admin, Barangay Official).

Local Developer Installation (Source Code)

1. Install Git

- a. Download and install Git from the official website:

2. Install Node.js

- a. Download and install the latest LTS version of Node.js from the official website

3. Clone the repository locally on your IDE using the web URL

- a. <https://github.com/flood-rescue-priority-dss/likas-app.git>

4. Navigate into the project directory

- a. `cd <insert_repository_folder_name>`
- b. Run `"pip install -r requirements.txt"`
- c. Run `"python -m uvicorn main:app --reload --port 8000"`

5. Navigate to the backend directory

- a. `cd likas-backend`

6. Install the required dependencies

- a. `npm install`

7. Install the required dependencies

- a. `npm install`

8. Launch the application in your browser:

- a. `npm run dev`

Deployment to Production

To deploy updates to the live production environment, administrators must follow this deployment pipeline:

Setup and Configuration:

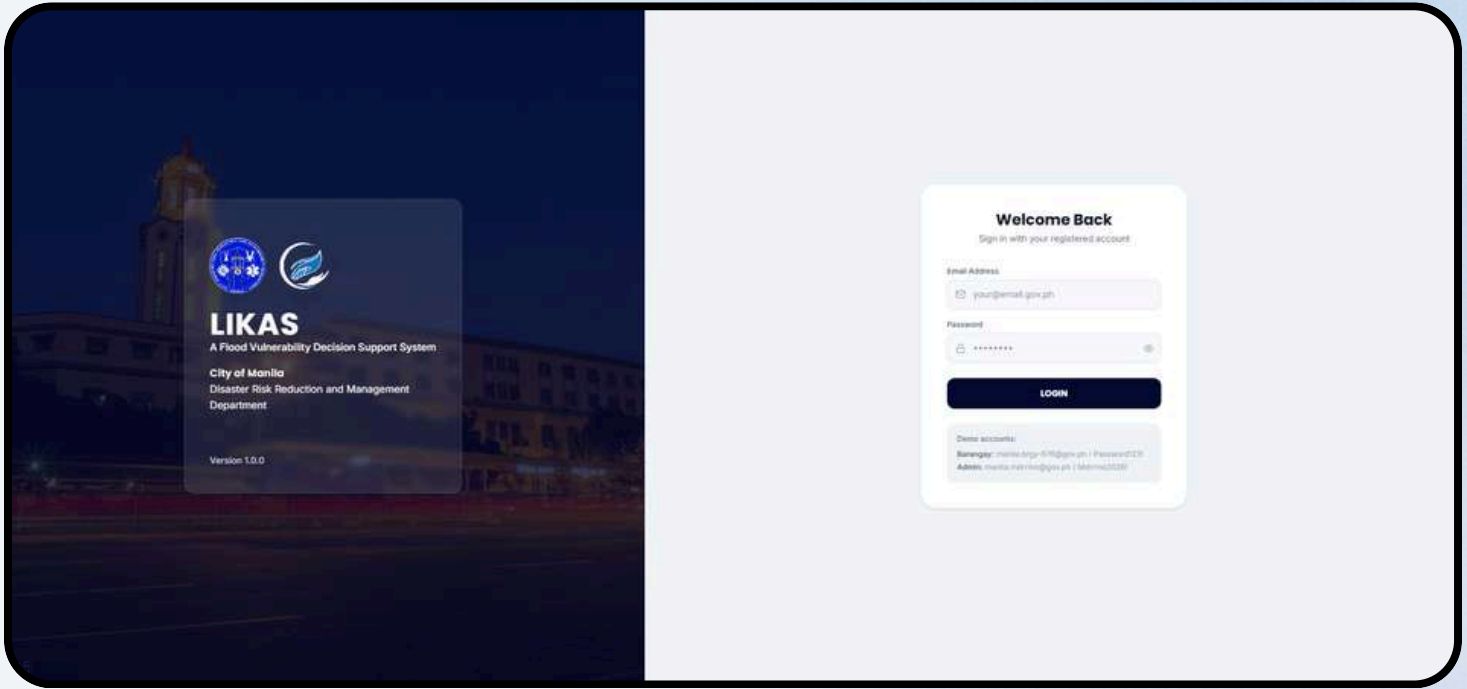
To deploy updates to the live production environment, administrators must follow this deployment pipeline:

1. **Version Control:** Push the finalized codebase to the main branch of the GitHub repository (flood-rescue-priority-dss/likas-app).
2. **Dependency Check:** Ensure each service's dependency manifest is up to date:
 - a. requirements.txt for the Python scoring microservice.
 - b. package.json for the Node.js backend and React frontend.
3. **Render Deployment:** Access the Render Dashboard using the authorized administrator account.
 - a. Python Microservice: Render automatically pulls the latest commit, installs dependencies via `pip install -r requirements.txt`, and starts the service using `uvicorn main:app --host 0.0.0.0 --port $PORT`.
 - b. Node.js Backend: Render builds the service using the project's Dockerfile and starts the container automatically.
 - c. React Frontend: Render runs `npm install && npm run build` and publishes the contents of the dist directory as a static site.
4. **Environment Variables:** Within each service's Environment tab, ensure all required variables (JWT_SECRET, DATABASE_URL, VITE_API_URL, etc.) are correctly configured before deployment.
5. **Go Live:** Once the build completes successfully, Render automatically deploys the update and propagates it to the live production URL.



LIKAS INTERFACE

Step-by-Step Usage



LOGIN

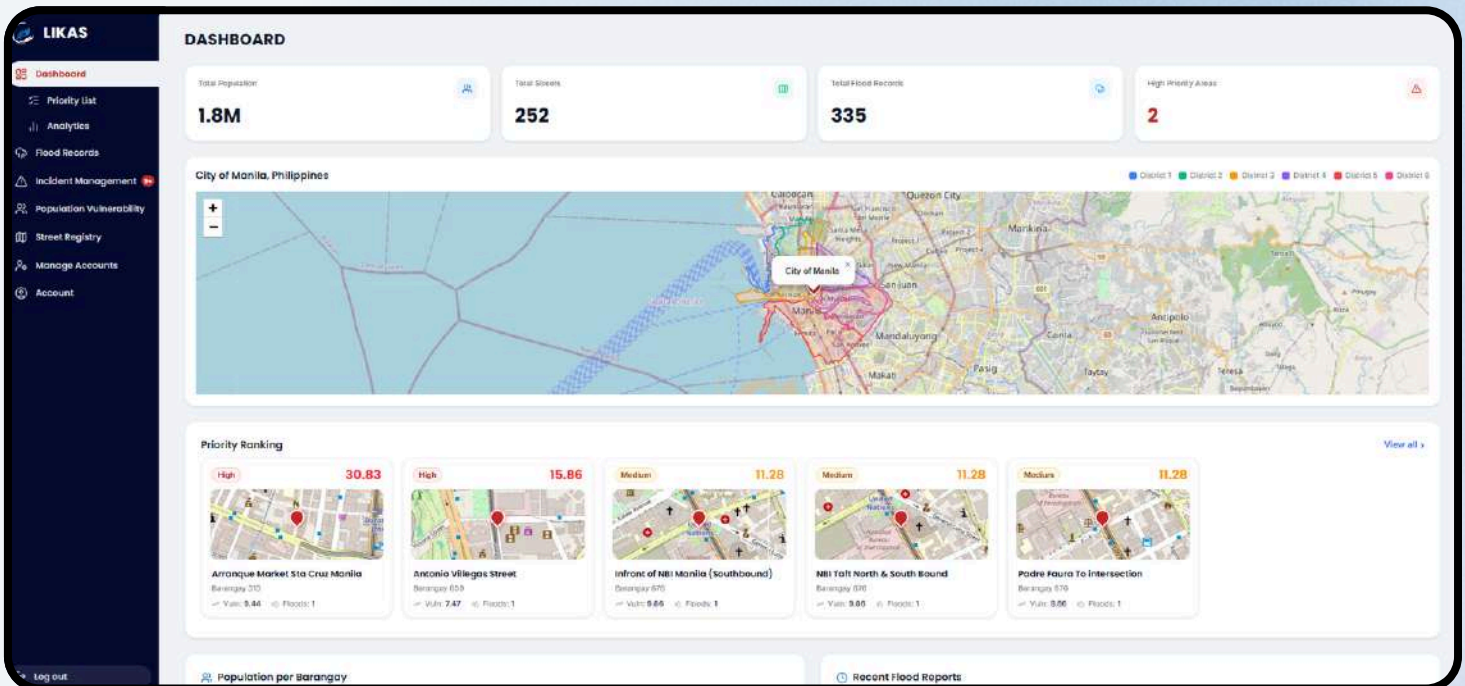
The Login Page allows authorized users to securely access the LIKAS system using their registered credentials.

1. Enter your registered Email Address.
2. Enter your Password.
3. Click the LOGIN button.
4. If the credentials are valid, the system will redirect you to the Dashboard.

NOTE !

Only registered users are allowed to access the system. An error message will be displayed if an invalid email address or password is entered.

Step-by-Step Usage – Admin



DASHBOARD

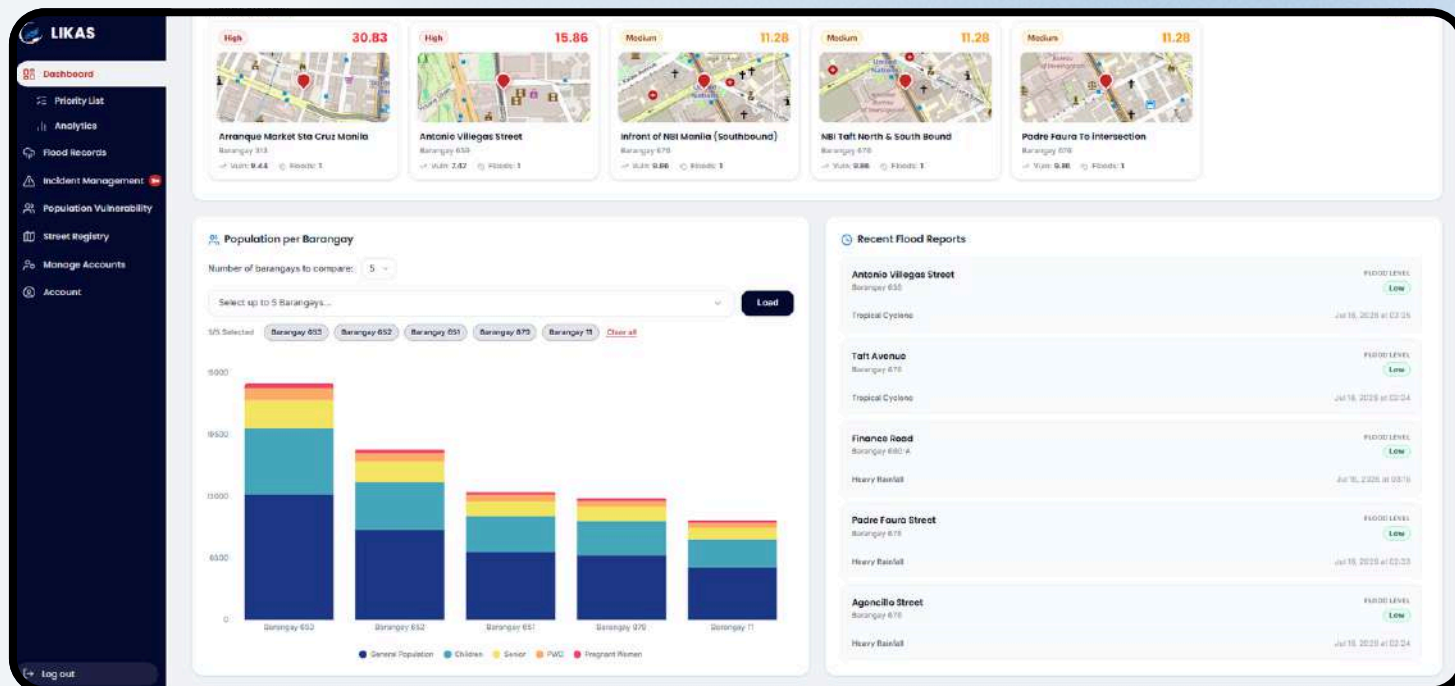
The Dashboard provides an overview of the current flood situation in the City of Manila. It displays key statistics, an interactive map, priority-ranked streets, population vulnerability data, and recent flood reports to help users monitor and assess flood-prone areas.

1. View the summary cards to monitor the total population, registered streets, recorded flood incidents, and identified high-priority areas.
2. Use the interactive map to view the locations and district boundaries within the City of Manila.
3. Review the Priority Ranking section to identify streets that require immediate attention based on their computed priority scores.

NOTE !

The Dashboard is automatically updated whenever new flood records, vulnerability data, or priority assessments are added to the system.

Step-by-Step Usage – Admin



DASHBOARD

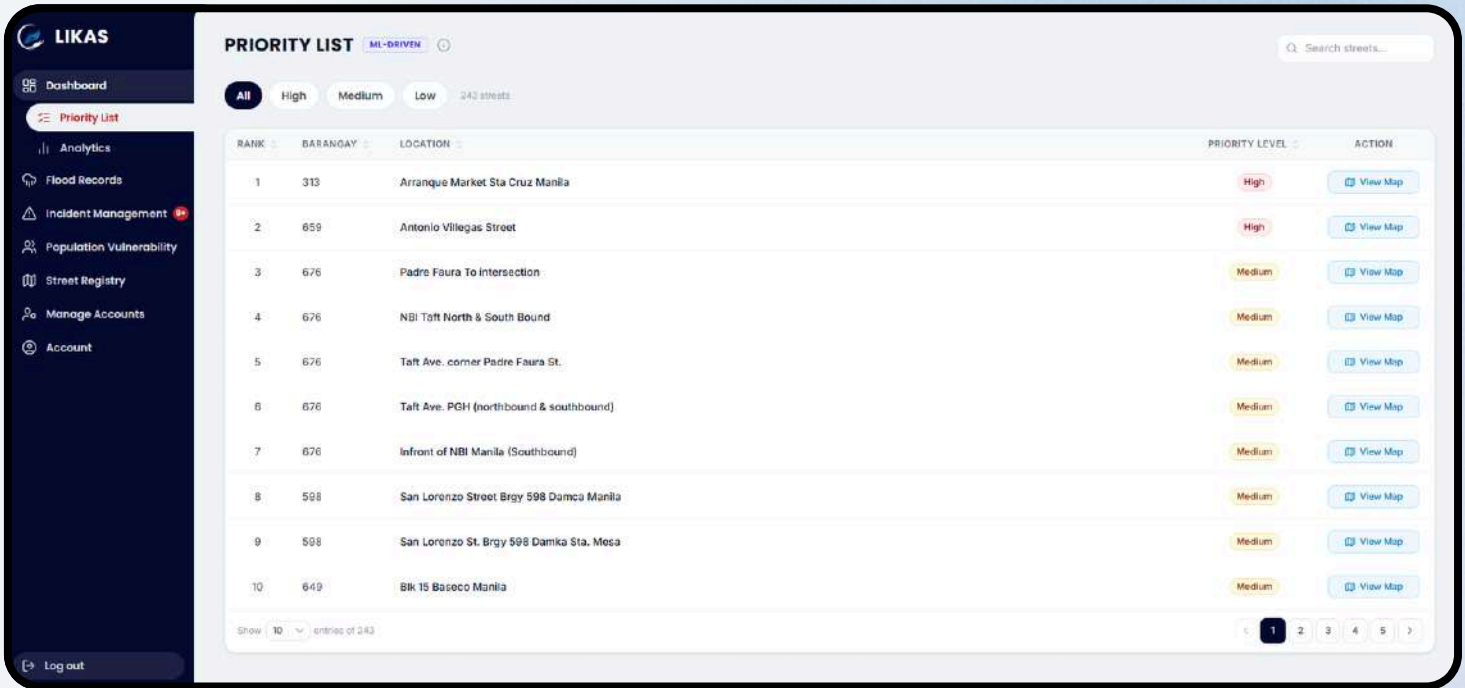
The Dashboard provides an overview of the current flood situation in the City of Manila. It displays key statistics, an interactive map, priority-ranked streets, population vulnerability data, and recent flood reports to help users monitor and assess flood-prone areas.

4. Compare the Population per Barangay by selecting up to five (5) barangays, then click Load to display the comparison chart.
5. Check the Recent Flood Reports panel to view the latest reported flood incidents, including their location, cause, flood level, and date and time of reporting.

NOTE !

The Dashboard is automatically updated whenever new flood records, vulnerability data, or priority assessments are added to the system.

Step-by-Step Usage – Admin

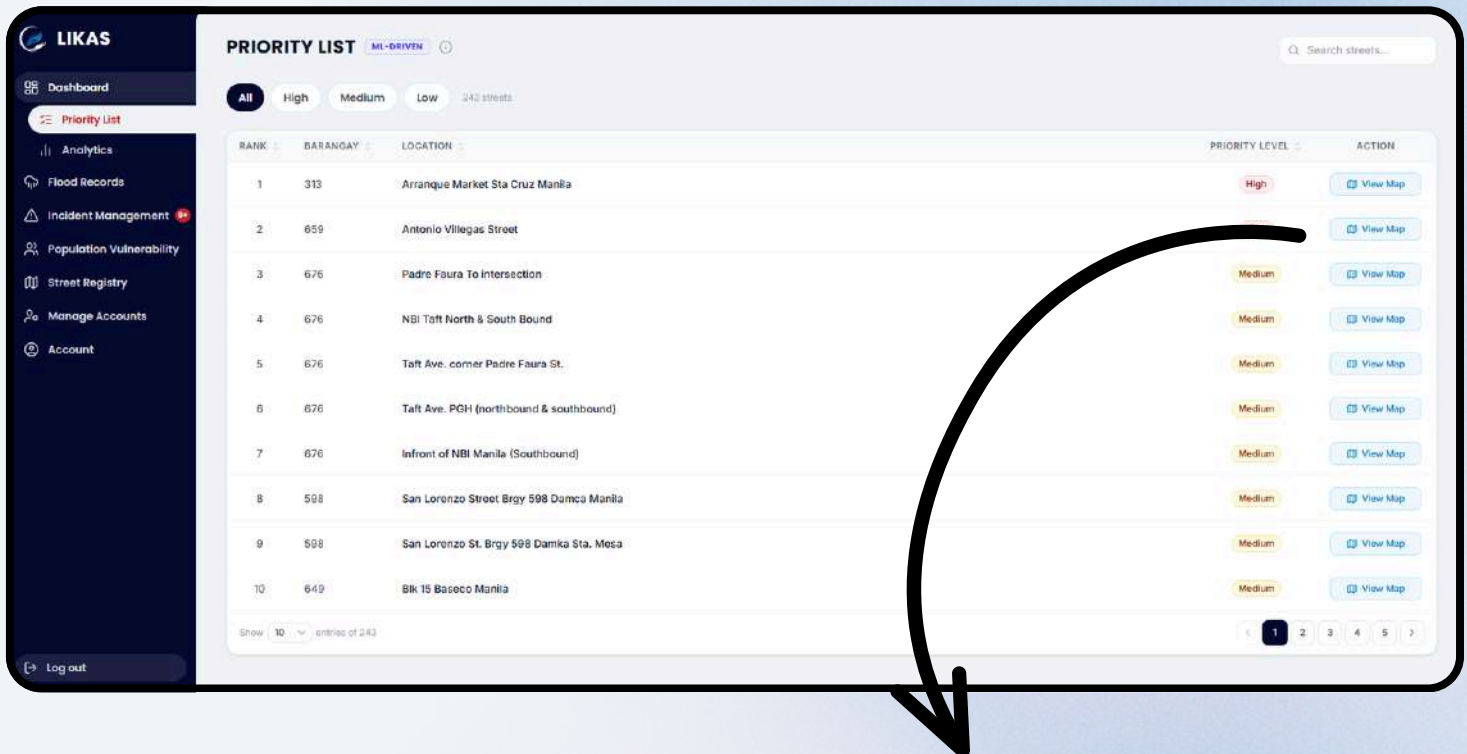


PRIORITY LIST

The Priority List displays all registered streets ranked according to their computed flood priority level using the system's machine learning model. This page helps users quickly identify areas that require immediate monitoring and response.

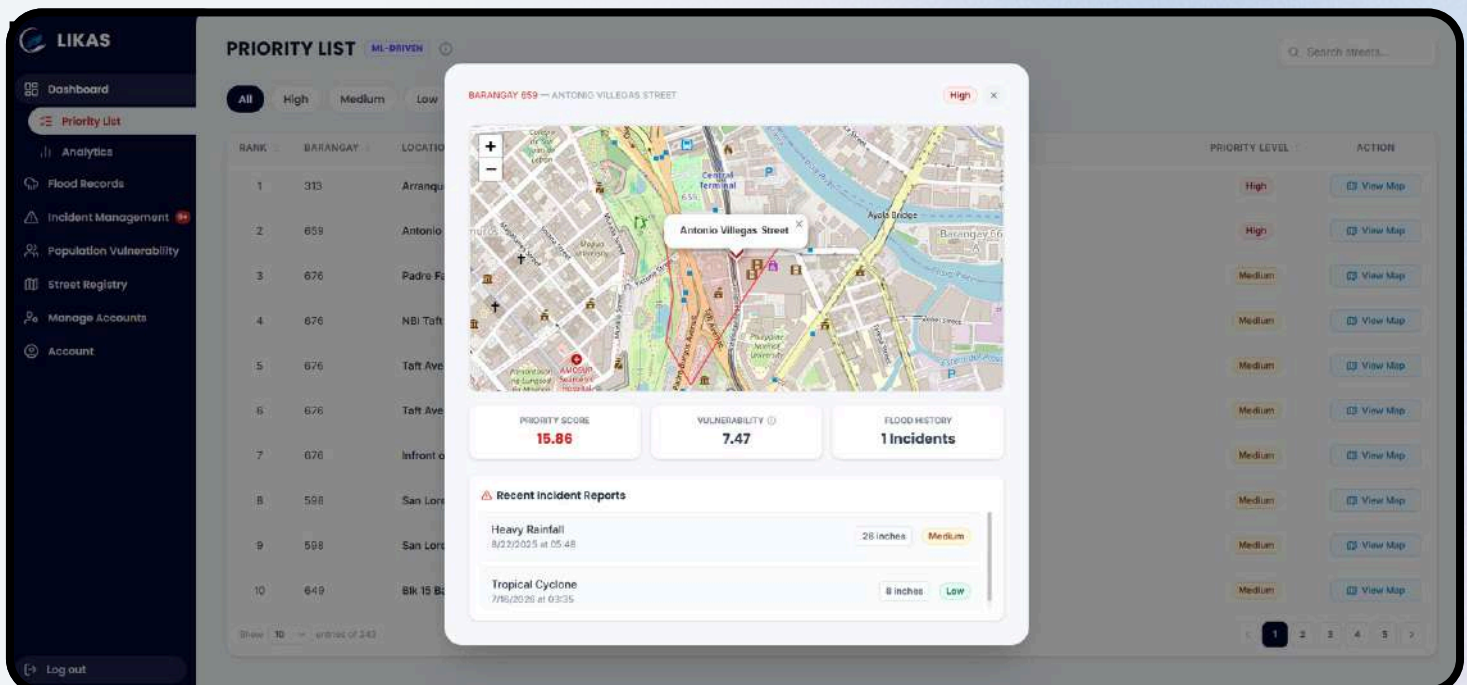
1. Use the Search bar to find a specific street.
2. Filter the list by selecting All, High, Medium, or Low priority levels.
3. Review the ranking, barangay, location, and corresponding priority level of each street.
4. Click View Map to display the selected street's location on the interactive map.
5. Use the pagination controls at the bottom of the page to navigate through additional records.

Step-by-Step Usage - Admin



The screenshot shows the LIKAS Priority List interface. On the left is a sidebar with navigation options: Dashboard, Priority List (selected), Analytics, Flood Records, Incident Management, Population Vulnerability, Street Registry, Manage Accounts, and Account. The main content area is titled 'PRIORITY LIST' and includes a search bar and filter tabs (All, High, Medium, Low). A table lists streets with columns for Rank, Barangay, Location, Priority Level, and Action. A large curved arrow points from the 'View Map' button for 'Antonio Villegas Street' to the second screenshot.

RANK	BARANGAY	LOCATION	PRIORITY LEVEL	ACTION
1	313	Arraque Market Sta Cruz Manila	High	View Map
2	659	Antonio Villegas Street	High	View Map
3	676	Padre Faura To Intersection	Medium	View Map
4	676	NBI Taft North & South Bound	Medium	View Map
5	676	Taft Ave. corner Padre Faura St.	Medium	View Map
6	676	Taft Ave. PGH (northbound & southbound)	Medium	View Map
7	676	Infront of NBI Manila (Southbound)	Medium	View Map
8	598	San Lorenzo Street Brgy 598 Damca Manila	Medium	View Map
9	598	San Lorenzo St. Brgy 598 Damca Sta. Mesa	Medium	View Map
10	649	Blk 15 Baseco Manila	Medium	View Map



This screenshot shows the same LIKAS Priority List interface, but with a detailed view of 'Antonio Villegas Street' (Barangay 659) overlaid. The view includes a map, a table of recent incident reports, and summary statistics for priority score, vulnerability, and flood history.

BARANGAY 659 — ANTONIO VILLEGAS STREET (High)

Summary:

- PRIORITY SCORE: 15.86
- VULNERABILITY: 7.47
- FLOOD HISTORY: 1 Incidents

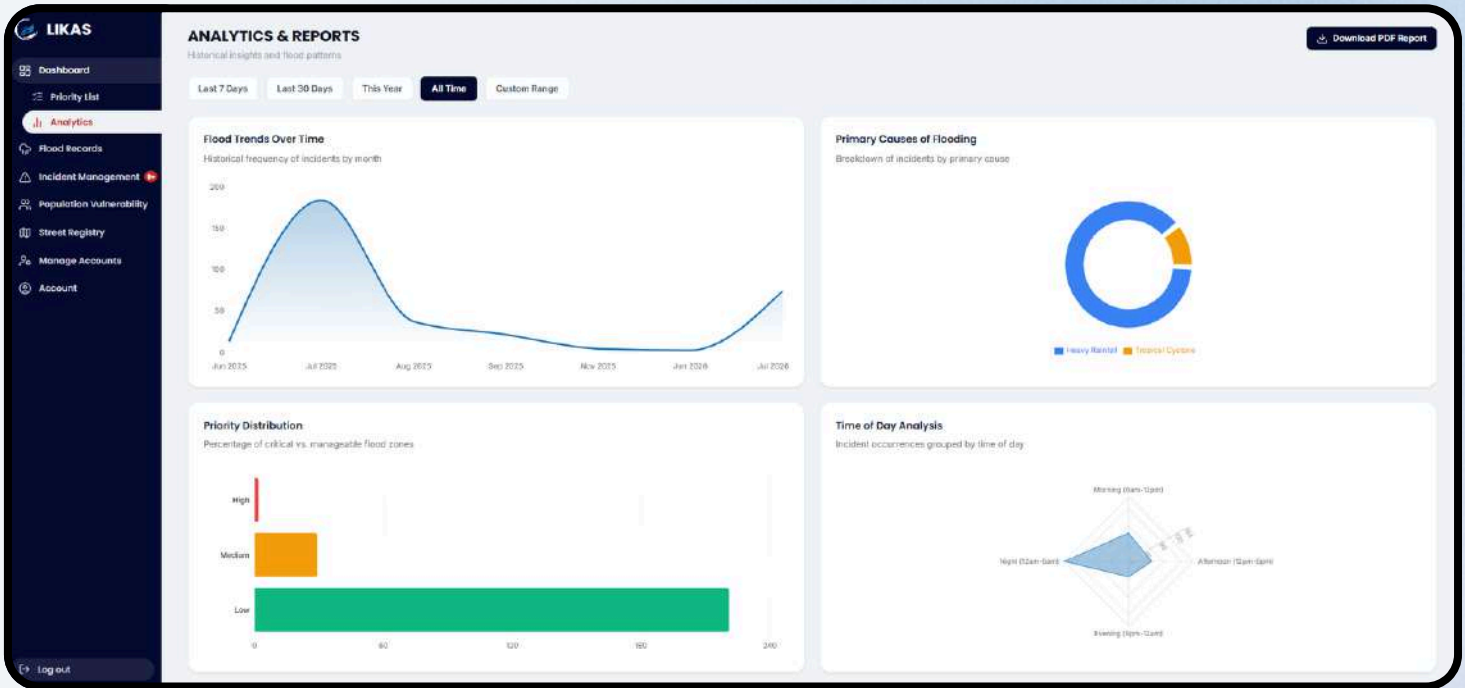
Recent Incident Reports:

Incident Type	Date/Time	Severity
Heavy Rainfall	8/22/2025 at 05:48	Medium
Tropical Cyclone	7/16/2026 at 03:35	Low

NOTE !

The Priority List is automatically updated whenever new flood records, vulnerability data, or priority assessments are processed by the system.

Step-by-Step Usage – Admin

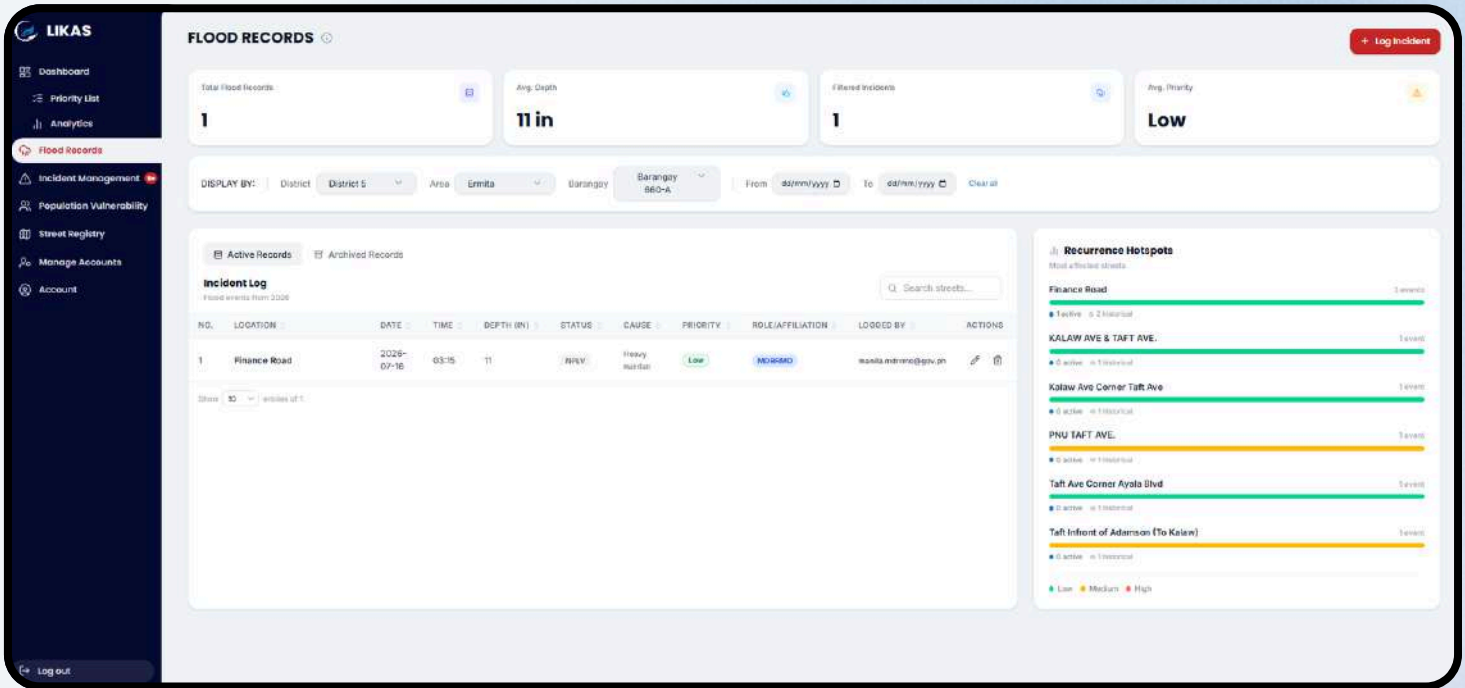


ANALYTICS & REPORTS

The Analytics page provides visual reports and historical insights on flood incidents within the City of Manila. It helps users analyze flood trends, identify common causes, monitor priority distributions, and understand incident patterns over time.

1. Select a date filter (Last 7 Days, Last 30 Days, This Year, All Time, or Custom Range) to display the desired data.
2. Review the Flood Trends Over Time chart to analyze the frequency of flood incidents.
3. View the Primary Causes of Flooding chart to identify the most common causes of reported floods.
4. Examine the Priority Distribution chart to compare the number of streets classified as High, Medium, or Low priority.
5. Check the Time of Day Analysis chart to determine when flood incidents most frequently occur.
6. Click Download PDF Report to generate and download a summary of the analytics for the selected period.

Step-by-Step Usage – Admin

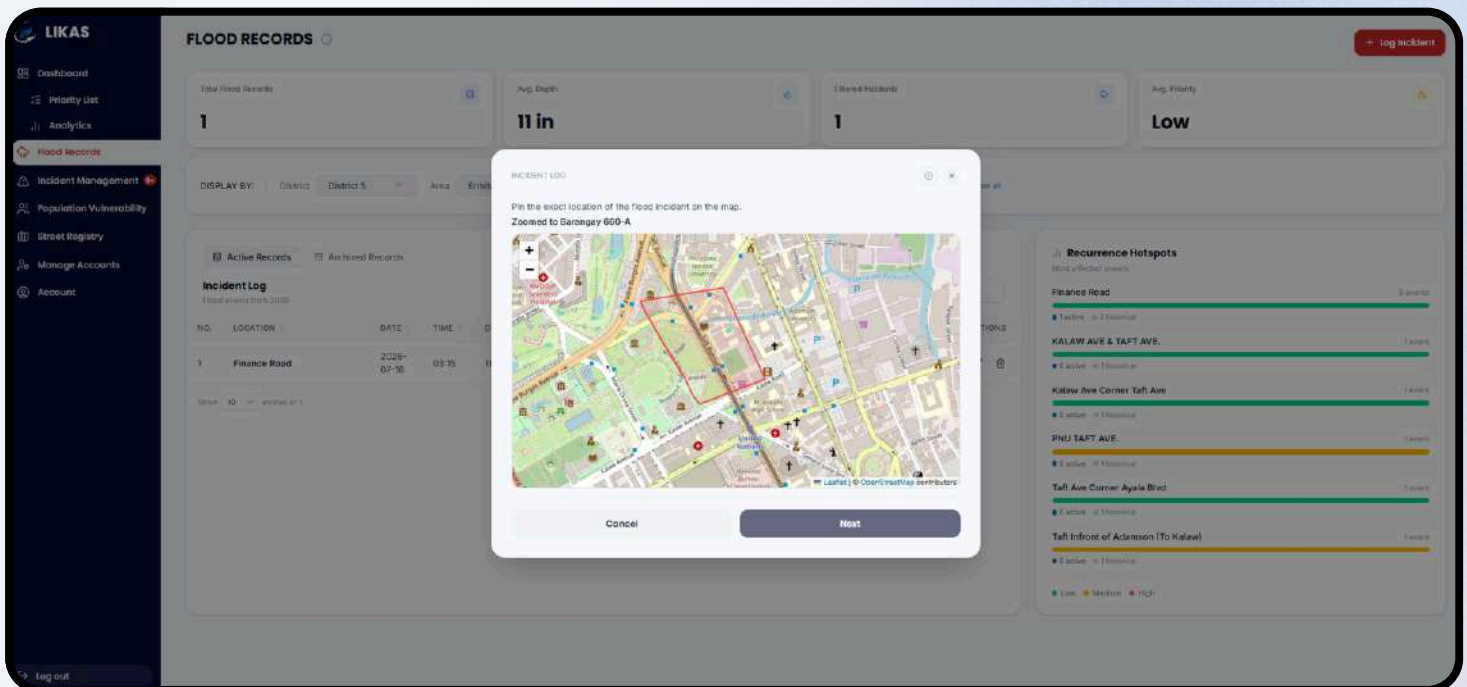
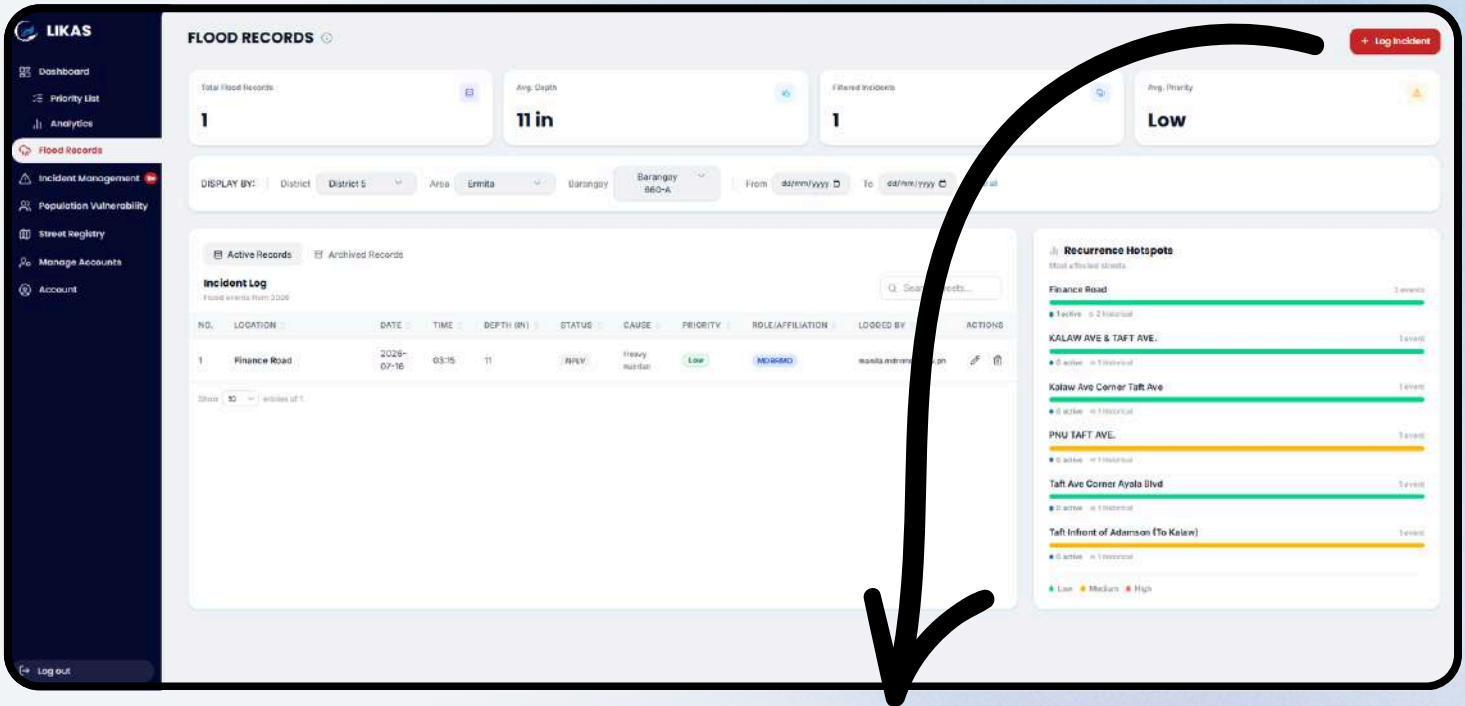


FLOOD RECORDS

The Flood Records page allows users to view, filter, and manage all recorded flood incidents within the system. It also provides summary statistics and identifies streets with recurring flood occurrences.

1. View the summary cards to monitor the total number of flood records, average flood depth, filtered incidents, and average priority level.
2. Filter flood records by District, Area, Barangay, or Date Range to display specific incidents.
3. Review the Incident Log to view the location, date, time, flood depth, status, cause, priority level, reporting office, and the user who logged the incident.
4. Use the Search bar to quickly locate a specific street or flood record.
5. Click the Edit or Delete icon under the Actions column to update or remove a flood record.
6. Select Archived Records to view previously archived flood incidents.
7. Click Log Incident to record a new flood incident.

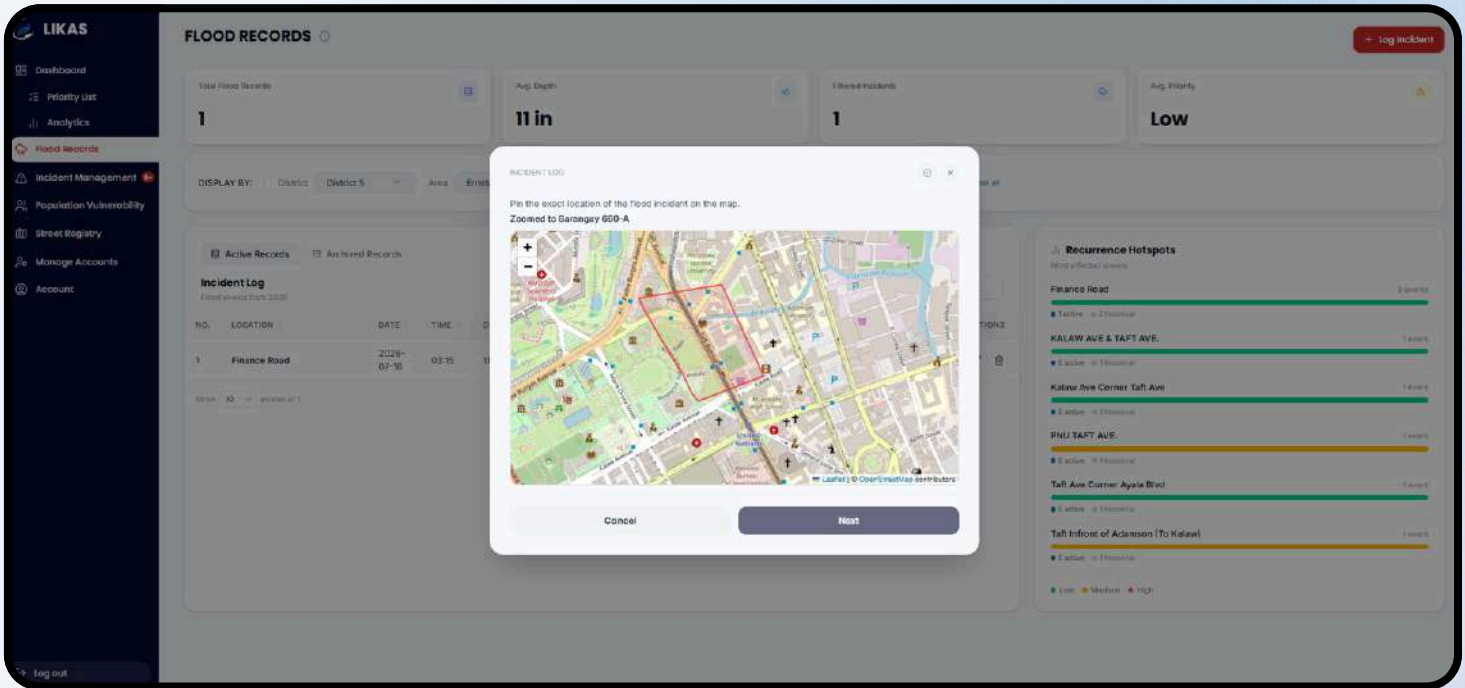
Step-by-Step Usage - Admin



NOTE !

The system automatically identifies the selected street from the pinned location and estimates the incident's priority level based on the entered flood depth.

Step-by-Step Usage – Admin



FLOOD RECORDS – LOG INCIDENT

The Log Incident feature allows authorized users to record a new flood incident within a selected barangay. The system validates the location before allowing the incident to be submitted.

1. Click Log Incident from the Flood Records page.
2. Pin the exact location of the flood incident on the interactive map.
3. Ensure that the selected location is within the highlighted barangay boundary. If the pin is placed outside the boundary, the system will display an "Outside barangay boundary" warning and prevent you from proceeding.
4. Once a valid location is selected, click Next.
5. Enter the required incident details, including:
 - a. Date
 - b. Time
 - c. Flood Depth
 - d. Cause of Flooding
 - e. Remarks (Upload an attachment)
6. Click Review Details to verify the entered information.
7. Review the incident summary, then click Confirm to save the flood incident.

Step-by-Step Usage – Admin

NO.	LOCATION	SUBMITTED BY	DATE	TIME	DEPTH (IN)	CAUSE	PRIORITY	LOGGED BY	STATUS	ACTIONS
1	Taft Avenue Barangay 676	Barangay 676	2026-07-16	03:34	8	Tropical Cyclone	Low	maria.brgy-676@gov.ph	Pending	View Reject Approve
2	Pedre Faura Street Barangay 676	Barangay 676	2026-07-16	02:33	8	Heavy Rainfall	Low	maria.brgy-676@gov.ph	Pending	View Reject Approve
3	Agoncillo Street Barangay 676	Barangay 676	2026-07-16	02:24	8	Heavy Rainfall	Low	maria.brgy-676@gov.ph	Pending	View Reject Approve
4	Leon Quinto Street Barangay 676	Barangay 676	2026-07-16	02:00	8	Heavy Rainfall	Low	maria.brgy-676@gov.ph	Pending	View Reject Approve
5	Pedre Faura Street Barangay 676	Barangay 676	2026-07-16	01:56	8	Heavy Rainfall	Low	maria.brgy-676@gov.ph	Pending	View Reject Approve
6	Pedre Faura Street Barangay 676	Barangay 676	2026-07-16	01:55	8	Heavy Rainfall	Low	maria.brgy-676@gov.ph	Pending	View Reject Approve
7	Pedre Faura Street Barangay 676	Barangay 676	2026-07-16	01:54	8	Heavy Rainfall	Low	maria.brgy-676@gov.ph	Pending	View Reject Approve
8	Taft Avenue Barangay 676	Barangay 676	2026-07-16	01:33	8	Heavy Rainfall	Low	maria.brgy-676@gov.ph	Pending	View Reject Approve

INCIDENT LOG MANAGEMENT

The Incident Management page allows administrators to review, verify, and manage submitted flood incident reports before they are added to the official flood records.

1. View the summary cards to monitor the number of Pending Review, Approved, and Rejected incident reports.
2. Select the Pending Approvals, Approved, or Rejected tab to view incident reports based on their verification status.
3. Use the Search bar to locate a specific street or incident report.
4. Click the View icon to review the complete details of a submitted incident.
5. Click Approve to verify and add the incident to the official flood records.
6. Click Reject to decline an invalid or incorrect incident submission.

NOTE !

Only administrators are authorized to approve or reject incident reports. Once approved, the incident is automatically added to the system's official flood records and reflected in the analytics and priority assessment.

Step-by-Step Usage – Admin

LIKAS

Dashboard

Priority List

Analytics

Flood Records

Incident Management

Population Vulnerability

Street Registry

Manage Accounts

Account

Log out

POPULATION VULNERABILITY

Total Population

1,846,513

Total PWD

93,231

Total Senior

221,371

Total Children

516,435

Total Pregnant

36,940

All Barangays

907 barangays

Search barangays...

NO.	BARANGAY	POPULATION	PWD	SENIOR	CHILDREN	PREGNANT	VULNERABILITY LEVEL	UPDATED	ACTION
1	Barangay 1	3,333	108	417	1,031	73	High		
2	Barangay 2	2,580	107	292	748	63	High		
3	Barangay 3	2,135	92	214	553	43	High		
4	Barangay 4	1,978	84	277	574	31	High		
5	Barangay 5	1,066	44	140	296	25	High		
6	Barangay 6	3,671	190	383	956	61	High		
7	Barangay 7	2,588	152	331	656	56	High		
8	Barangay 8	4,345	252	533	1,338	95	High		
9	Barangay 9	3,576	208	398	1,044	74	High		
10	Barangay 10	2,216	101	304	685	47	High		

Show 30 entries of 907

1

2

3

4

5

POPULATION VULNERABILITY

The Population Vulnerability page displays demographic information for each barangay, including vulnerable population groups used in the system's priority assessment.

1. View the summary cards to monitor the total population, persons with disabilities (PWDs), senior citizens, children, and pregnant women.
2. Review the barangay list to view the population and vulnerable group counts for each barangay.
3. Use the Search bar to quickly locate a specific barangay.
4. Check the Vulnerability Level assigned to each barangay based on its demographic data.
5. Click the Edit icon under the Action column to update the population and vulnerability information of a barangay.

NOTE !

Changes made to a barangay's vulnerability data are automatically considered in the system's priority assessment and ranking process.

Step-by-Step Usage – Admin

STREET REGISTRY

Total Streets: 252 | Avg. Priority Score: 4 | Avg. Vulnerability Score: 9 | High Priority Streets: 2

DISPLAY BY: District: All Districts | Area: All Areas | Barangay: All Barangays

All Locations (252 available locations) Search streets...

ID	LOCATION	FLOOD HAZARD	VULNERABILITY	PRIORITY	LAST UPDATED
1	Arraquee Market Sta Cruz Manila	High	High	High	2026-07-19
2	Antonio Villegas Street	High	Medium	High	2026-07-19
3	Taft Ave. corner Padre Faura St.	High	High	Medium	2026-07-19
4	NBI Taft North & South Bound	High	High	Medium	2026-07-19
5	Padre Faura To Intersection	High	High	Medium	2026-07-19
6	Infront of NBI Manila (Southbound)	High	High	Medium	2026-07-19
7	Taft Ave. PCH (northbound & southbound)	High	High	Medium	2026-07-19
8	San Lorenzo Street Brgy 598 Damica Manila	High	High	Medium	2026-07-19
9	San Lorenzo St. Brgy 598 Damica Sta. Mesa	High	High	Medium	2026-07-19
10	Blk 1 Dubai Baseco	High	High	Medium	2026-07-19

Show 10 entries of 202

Map Preview: A map showing the location of the selected street (Blk 1 Dubai Baseco) in Manila.

STREET REGISTRY

The Street Registry page allows users to view and monitor all registered street locations, including their flood hazard, vulnerability, and priority classifications.

1. View the summary cards to monitor the total number of streets, average priority score, average vulnerability score, and high-priority streets.
2. Use the District, Area, and Barangay filters to display specific street locations.
3. Browse the list of registered streets and review their Flood Hazard, Vulnerability, and Priority levels.
4. Use the Search bar to quickly locate a specific street or location.
5. Click a location from the list to highlight and preview its position on the map.

NOTE !

Street classifications are automatically updated based on recorded flood incidents and population vulnerability data used in the system's priority assessment process.

Step-by-Step Usage - Admin

The screenshot shows the LIKAS Street Registry dashboard. On the left is a sidebar with navigation links: Dashboard, Priority List, Analytics, Flood Records, Incident Management, Population Vulnerability, Street Registry (highlighted), Manage Accounts, and Account. The main content area is titled 'STREET REGISTRY' and displays summary statistics: Total Streets (252), Avg. Priority Score (4), Avg. Vulnerability Score (9), and High Priority Streets (2). Below these are filters for 'DISPLAY BY' (District, All Districts, Area, All Areas, Barangay, All Barangays) and a search bar. A table titled 'All Locations' lists 10 entries with columns for NO., LOCATION, FLOOD HAZARD, VULNERABILITY, PRIORITY, and LAST UPDATED. The table shows various streets in Manila with their respective hazard and vulnerability levels. To the right of the table is a 'Map Preview' showing a map of Manila with a red pin indicating the location of the selected street.

NO.	LOCATION	FLOOD HAZARD	VULNERABILITY	PRIORITY	LAST UPDATED
1	Arraque Market Sta Cruz Manila	High	High	High	2026-07-09
2	Antonio Villegas Street	High	Medium	High	2026-07-09
3	Taft Ave. corner Padre Faura St.	High	High	Medium	2026-07-09
4	NBI Taft North & South Bound	High	High	Medium	2026-07-09
5	Padre Faura To Intersection	High	High	Medium	2026-07-09
6	Infront of NBI Manila (Southbound)	High	High	Medium	2026-07-09
7	Taft Ave. PGH (northbound & southbound)	High	High	Medium	2026-07-09
8	San Lorenzo Street Brgy 598 Damca Manila	High	High	Medium	2026-07-09
9	San Lorenzo St. Brgy 598 Damca Sta. Mesa	High	High	Medium	2026-07-09
10	Blk 1 Dubai Baseco	High	High	Medium	2026-07-09

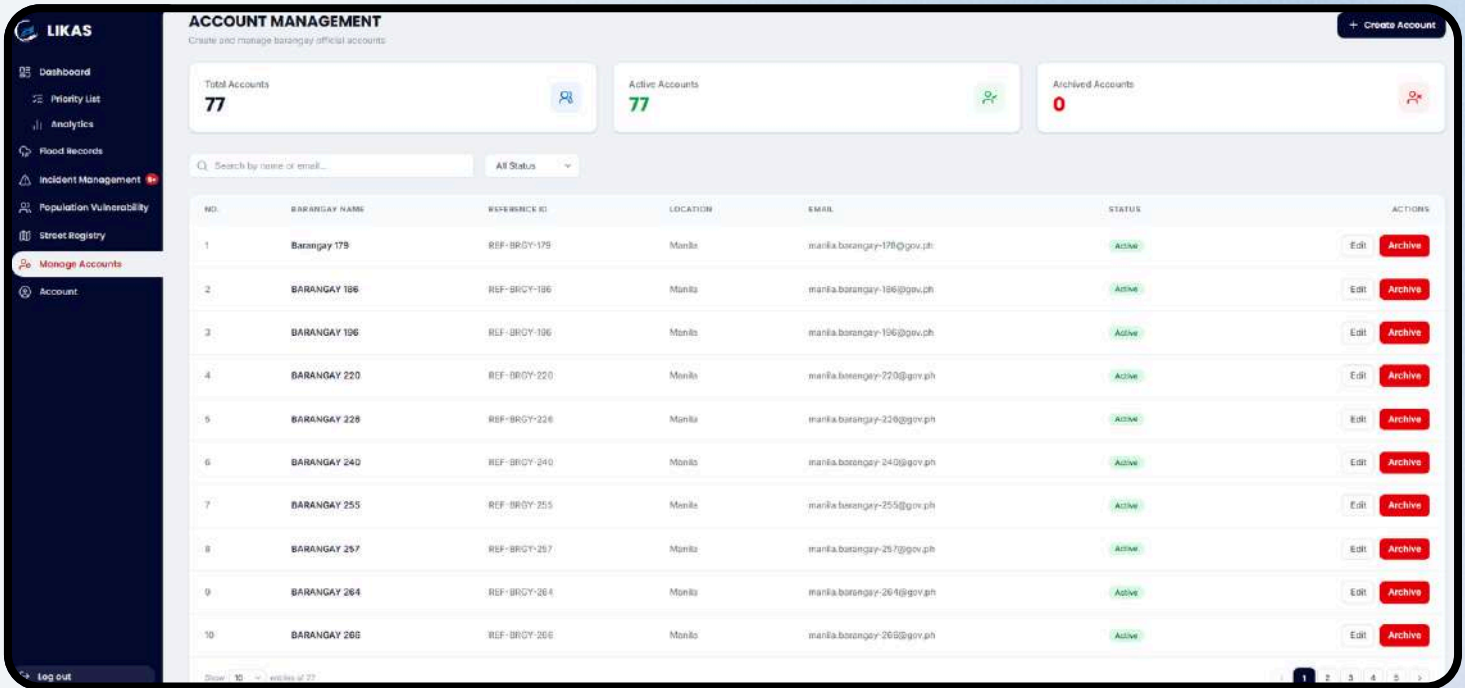
This screenshot shows the same LIKAS Street Registry dashboard, but with the row for 'Padre Faura To Intersection' selected. The table now shows 10 entries, with the selected row highlighted. The 'Map Preview' on the right has been updated to show a detailed map of the area around the selected location, with a red pin indicating the specific intersection. The map also shows surrounding streets and landmarks.

NO.	LOCATION	FLOOD HAZARD	VULNERABILITY	PRIORITY	LAST UPDATED
1	Arraque Market Sta Cruz Manila	High	High	High	2026-07-10
2	Antonio Villegas Street	High	Medium	High	2026-07-10
3	Taft Ave. corner Padre Faura St.	High	High	Medium	2026-07-10
4	NBI Taft North & South Bound	High	High	Medium	2026-07-10
5	Padre Faura To Intersection	High	High	Medium	2026-07-10
6	Infront of NBI Manila (Southbound)	High	High	Medium	2026-07-10
7	Taft Ave. PGH (northbound & southbound)	High	High	Medium	2026-07-10
8	San Lorenzo Street Brgy 598 Damca Manila	High	High	Medium	2026-07-10
9	San Lorenzo St. Brgy 598 Damca Sta. Mesa	High	High	Medium	2026-07-10
10	Blk 1 Dubai Baseco	High	High	Medium	2026-07-10

NOTE !

The map preview automatically updates when a street row is selected.

Step-by-Step Usage – Admin



MANAGE ACCOUNTS

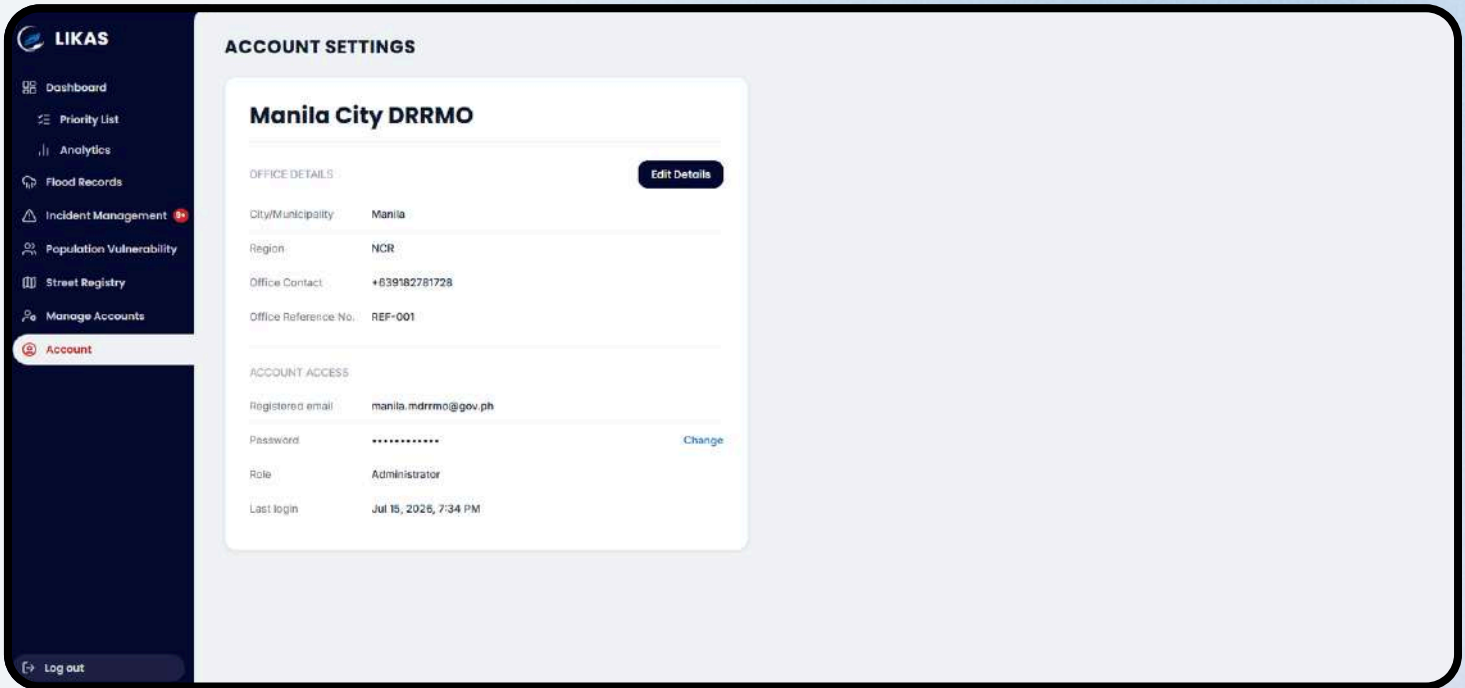
The Account Management page allows admins to create, search, edit, archive, reactivate, and manage user accounts.

1. To generate a new user, click "Create Account". Fill in details and authorize with Admin password.
2. Find a specific user by using search function. Type in any of the following user information and allow the system to automatically filter the results as you type and match your query (e.g., Juan, manila.drrmo, tondo or TONDO)
3. Entries may be filtered by all, active, and archived records. Therefore, admins can decide whether an account is active or archived.
4. To restore an account, toggle to Archived view, and click Reactivate.

NOTE

To edit account details, click Edit button, modify existing information, and save changes. The page is strictly authorized to system administrators (MDRRMD) and hidden for Barangay users .

Step-by-Step Usage – Admin



ACCOUNT SETTINGS

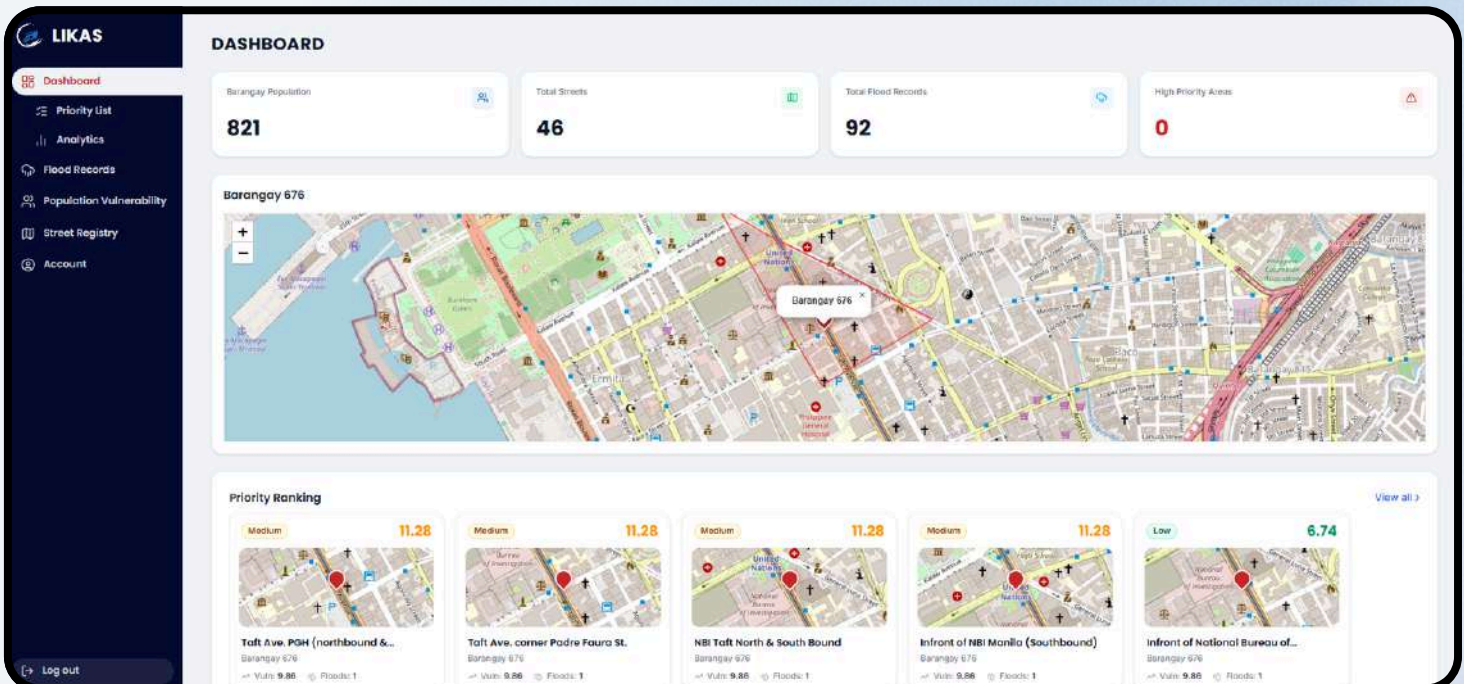
The Account Settings page allows administrators to view and manage office information and account credentials used in the system.

1. Review the Office Details section, including the city/municipality, region, office contact number, and office reference number.
2. Click Edit Details to update the office information displayed in the system.
3. View the Account Access section to check the registered email address, role, and last login information.
4. Click Change beside the password field to update your account password.

NOTE !

Only authorized account holders can modify account details and credentials. Changes made to office information are reflected throughout the system where applicable.

Step-by-Step Usage – BARANGAY



DASHBOARD

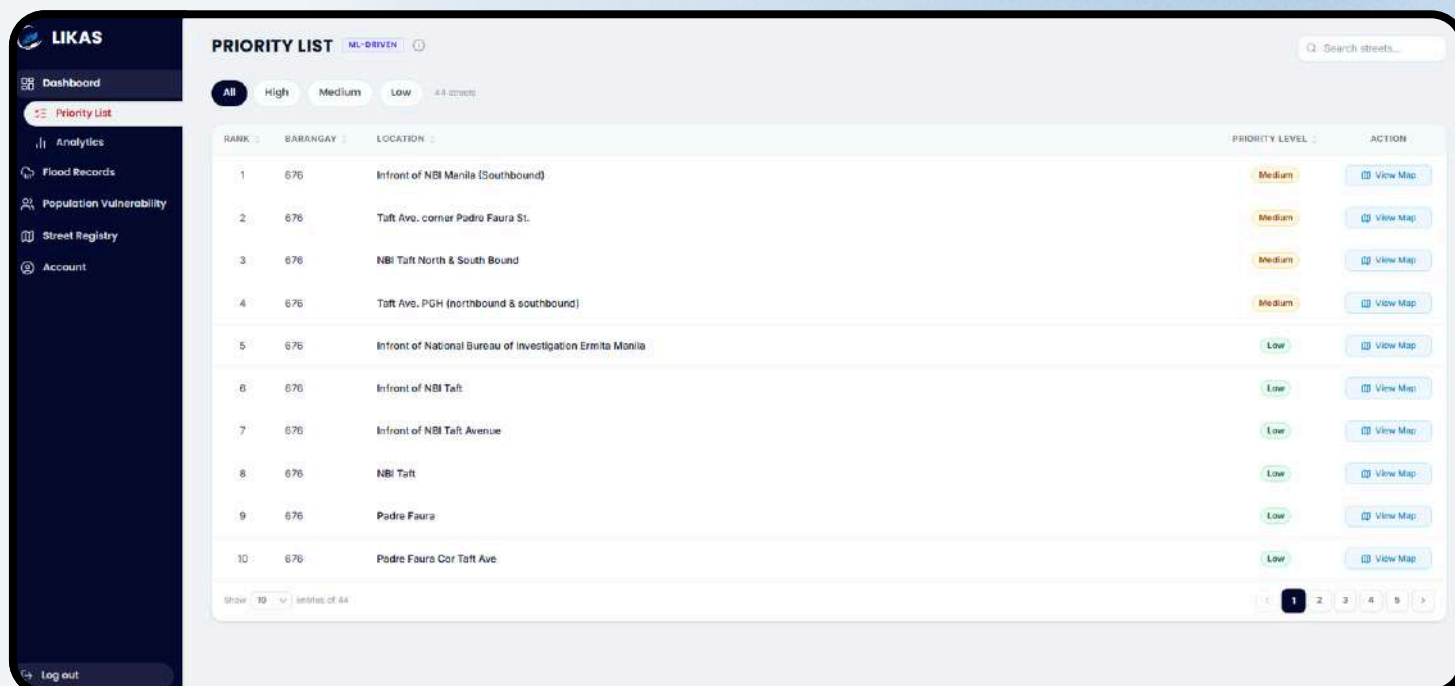
The Dashboard provides an overview of the selected barangay's flood-related information, including population data, flood records, and priority rankings.

1. View the summary cards to monitor the barangay population, total registered streets, total flood records, and high-priority areas.
2. Review the interactive map to view the selected barangay boundary and location.
3. Explore the Priority Ranking section to identify streets based on their calculated priority scores.
4. Review each ranked street's priority level, vulnerability score, and recorded flood incidents.
5. Click View All to access the complete priority ranking list for the selected barangay.

NOTE !

The dashboard automatically updates based on approved flood records and population vulnerability data.

Step-by-Step Usage – BARANGAY



PRIORITY LIST

The Priority List page displays streets ranked according to the system's machine learning–based priority assessment, helping users identify locations that may require attention and monitoring.

1. View the ranked list of streets based on their calculated priority scores.
2. Use the All, High, Medium, and Low filters to display streets by priority level.
3. Use the Search bar to quickly locate a specific street or location.
4. Review the Priority Level assigned to each street.
5. Click View Map to display the selected street's location on the map.

NOTE !

Rankings are updated whenever new flood records or vulnerability information are added to the system.

Step-by-Step Usage – BARANGAY

LIKAS

Dashboard
Priority List
Analytics
Flood Records
Population Vulnerability
Street Registry
Account
Log out

PRIORITY LIST ML-DRIVEN Search streets...

All High Medium Low 44 streets

RANK	BARANGAY	LOCATION	PRIORITY LEVEL	ACTION
1	676	Infront of NBI Manila (Southbound)	Medium	View Map
2	676	Taft Ave. corner Padre Faura St.	Medium	View Map
3	676	NBI Taft North & South Bound	Medium	View Map
4	676	Taft Ave. PGH (northbound & southbound)	Medium	View Map
5	676	Infront of National Bureau of Investigation Ermita Manila	Low	View Map
6	676	Infront of NBI Taft	Low	View Map
7	676	Infront of NBI Taft Avenue	Low	View Map
8	676	NBI Taft	Low	View Map
9	676	Padre Faura	Low	View Map
10	676	Padre Faura Cor Taft Ave	Low	View Map

Show 10 entries of 44

LIKAS

Dashboard
Priority List
Analytics
Flood Records
Population Vulnerability
Street Registry
Account
Log out

PRIORITY LIST ML-DRIVEN Search streets...

All High Medium Low 44 streets

RANK	BARANGAY	LOCATION	PRIORITY LEVEL	ACTION
1	676	Infront of NBI Manila (Southbound)	Medium	View Map
2	676	Taft Ave. corner Padre Faura St.	Medium	View Map
3	676	NBI Taft North & South Bound	Medium	View Map
4	676	Taft Ave. PGH (northbound & southbound)	Medium	View Map
5	676	Infront of National Bureau of Investigation Ermita Manila	Low	View Map
6	676	Infront of NBI Taft	Low	View Map
7	676	Infront of NBI Taft Avenue	Low	View Map
8	676	NBI Taft	Low	View Map
9	676	Padre Faura	Low	View Map
10	676	Padre Faura Cor Taft Ave	Low	View Map

Show 10 entries of 44

BARANGAY 676 — NBI TAFT NORTH & SOUTH BOUND Medium

Map View: NBI Taft North & South Bound

PRIORITY SCORE 11.28

VULNERABILITY 9.86

FLOOD HISTORY 1 Incidents

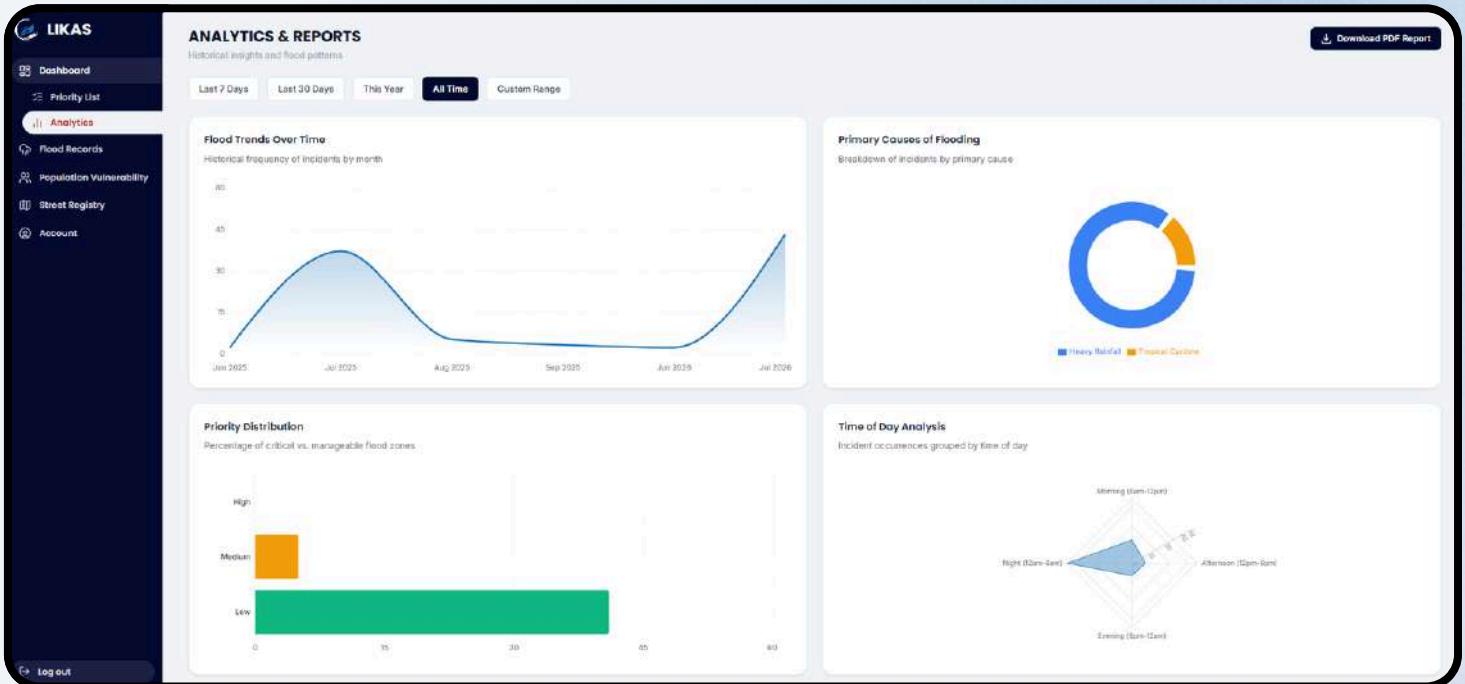
Recent Incident Reports

Heavy Rainfall
7/21/2023 at 0:20
19 inches Medium

NOTE !

The Priority List is automatically updated whenever new flood records, vulnerability data, or priority assessments are processed by the system.

Step-by-Step Usage – BARANGAY

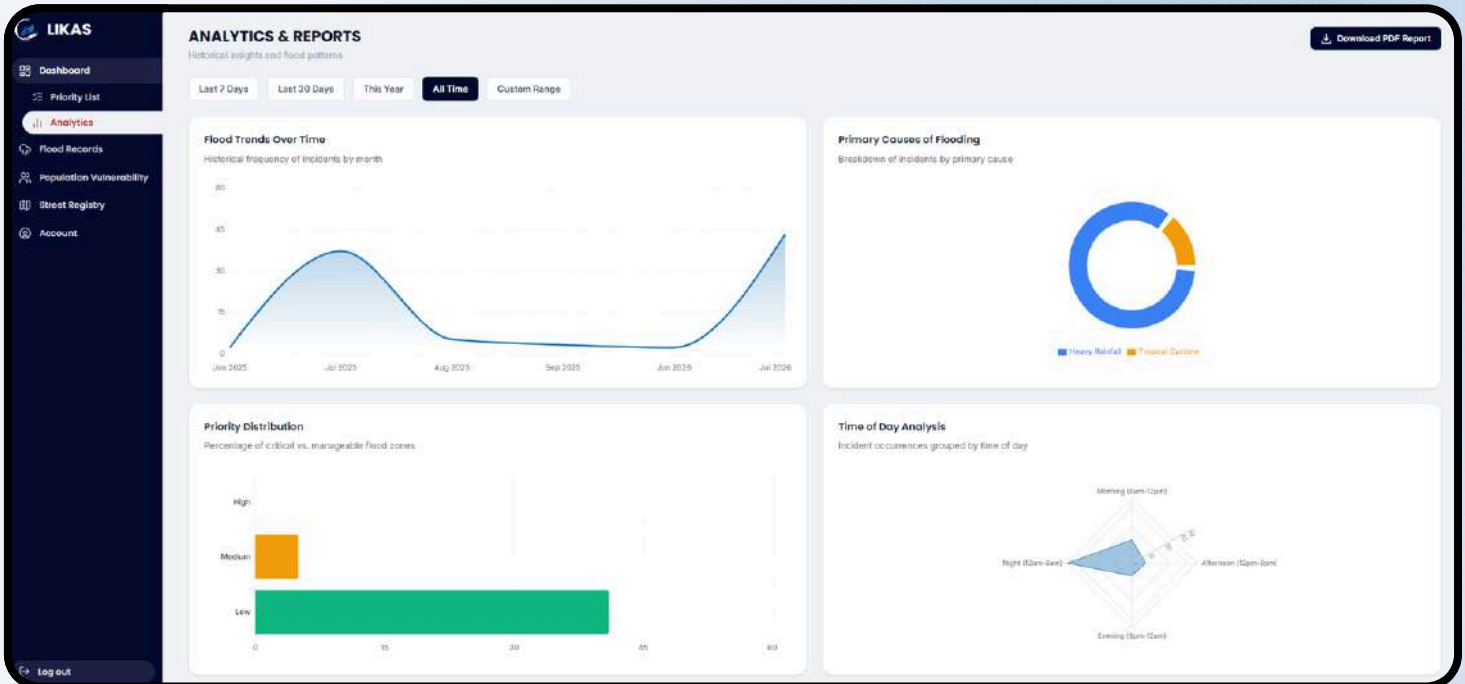


ANALYTICS & REPORTS

The Analytics & Reports page provides visual insights and statistical summaries of recorded flood incidents to support monitoring and decision-making.

1. Use the available date filters to view analytics for the Last 7 Days, Last 30 Days, This Year, All Time, or a Custom Range.
2. Review the Flood Trends Over Time chart to analyze the frequency of flood incidents across different periods.
3. Examine the Primary Causes of Flooding chart to identify the most common causes of recorded incidents.
4. View the Priority Distribution chart to compare the number of streets across different priority levels.
5. Analyze the Time of Day Analysis chart to determine when flood incidents most frequently occur.
6. Click Download PDF Report to generate and export the displayed analytics and reports.

Step-by-Step Usage – BARANGAY



FLOOD RECORDS

The Flood Records page allows Barangay users to view, filter, and manage flood incidents within their assigned barangay. It also provides summary statistics and enables users to record and update flood incident information.

1. View the summary cards to monitor the total number of flood records, average flood depth, filtered incidents, and average priority level.
2. Filter flood records by Area, Street, Status, or Date Range to display specific incidents within the barangay.
3. Review the Incident Log to view the incident location, date and time, flood depth, status, cause, priority level, reporting office, and the user who recorded the incident.
4. Use the Search bar to quickly locate a specific flood incident or street record.
5. Click the View, Edit or Delete icon under the Actions column to update or remove a flood record.
6. Click Log Incident to record a new flood incident within the barangay.
7. Review the updated flood records and summary statistics after creating or editing an incident.

Step-by-Step Usage - BARANGAY

The screenshot displays the LIKAS FLOOD RECORDS dashboard. On the left is a sidebar with navigation links: Dashboard, Priority List, Analytics, Flood Records (active), Population Vulnerability, Street Registry, and Account. The main content area is titled 'FLOOD RECORDS' and includes a '+ Log Incident' button. It features summary cards for 'Total Flood Records' (45), 'Avg. Depth' (11 in), 'Filtered Incidents' (45), and 'Avg. Priority' (Low). Below these is a 'TIME INTERVAL' selector. The 'Incident Log' section shows a table of flood events with columns for NO., LOCATION, DATE, TIME, DEPTH (IN), STATUS, CAUSE, PRIORITY, ROLE/AFFILIATION, and LOGGED BY. The table lists 6 incidents, all occurring on 2026-07-16, with locations like Taft Avenue, Padre Faura Street, and Agoncillo Street. To the right, the 'Recurrence Hotspots' chart shows the most affected streets with horizontal bar charts for each, indicating the number of events and their frequency.

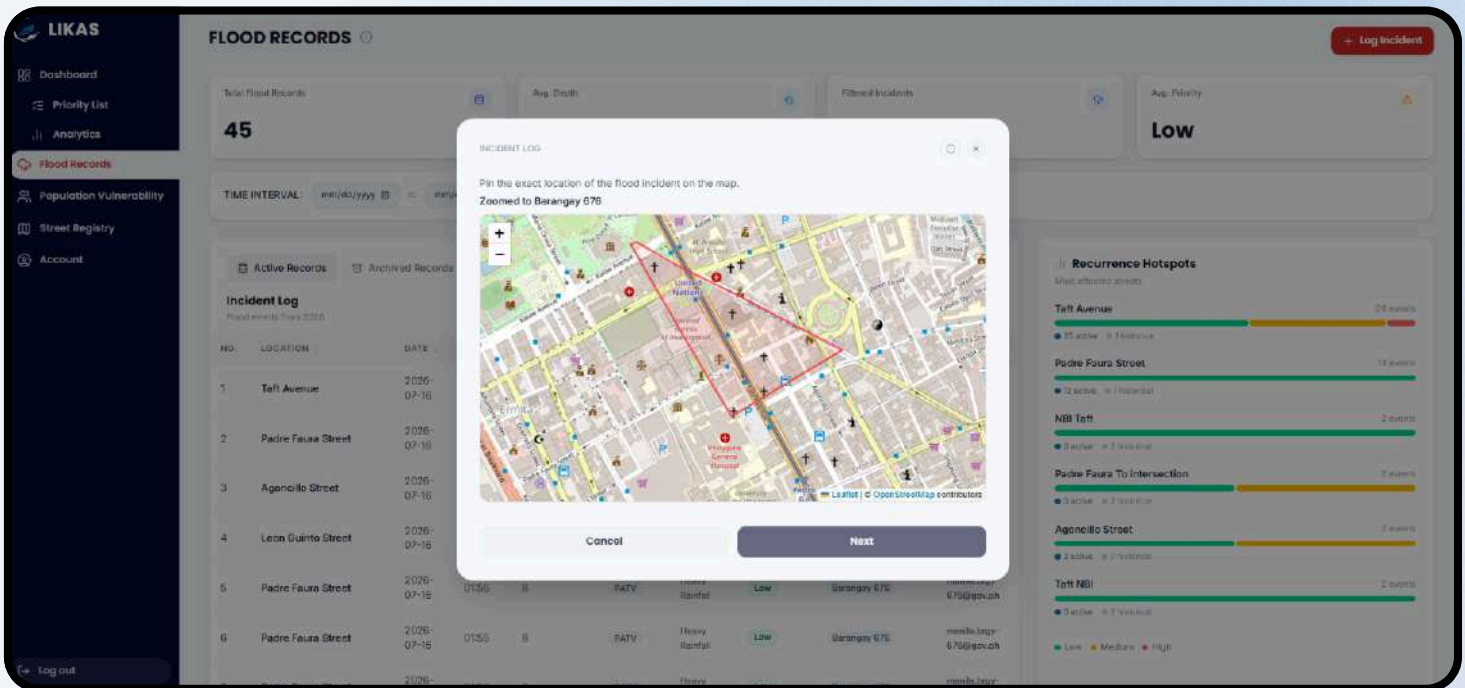
NO.	LOCATION	DATE	TIME	DEPTH (IN)	STATUS	CAUSE	PRIORITY	ROLE/AFFILIATION	LOGGED BY
1	Taft Avenue	2026-07-16	03:34	8	PATV	Tropical Cyclone	Low	Barangay 676	manila.brgy-676@gov.ph
2	Padre Faura Street	2026-07-16	02:33	8	PATV	Heavy Rainfall	Low	Barangay 676	manila.brgy-676@gov.ph
3	Agoncillo Street	2026-07-16	02:24	8	PATV	Heavy Rainfall	Low	Barangay 676	manila.brgy-676@gov.ph
4	Leon Guinto Street	2026-07-16	02:00	8	PATV	Heavy Rainfall	Low	Barangay 676	manila.brgy-676@gov.ph
5	Padre Faura Street	2026-07-16	01:56	8	PATV	Heavy Rainfall	Low	Barangay 676	manila.brgy-676@gov.ph
6	Padre Faura Street	2026-07-16	01:55	8	PATV	Heavy Rainfall	Low	Barangay 676	manila.brgy-676@gov.ph

This screenshot shows the same LIKAS FLOOD RECORDS dashboard, but with an 'INCIDENT LOG' modal open in the center. The modal prompts the user to 'Pin the exact location of the flood incident on the map' and notes it is 'Zoomed to Barangay 676'. It contains a map of the area with a red pin and a red bounding box. Below the map are 'Cancel' and 'Next' buttons. The background dashboard is dimmed, showing the same summary cards and incident log table as the previous screenshot.

NOTE !

The system automatically identifies the selected street from the pinned location and

Step-by-Step Usage – BARANGAY



FLOOD RECORDS – LOG INCIDENT

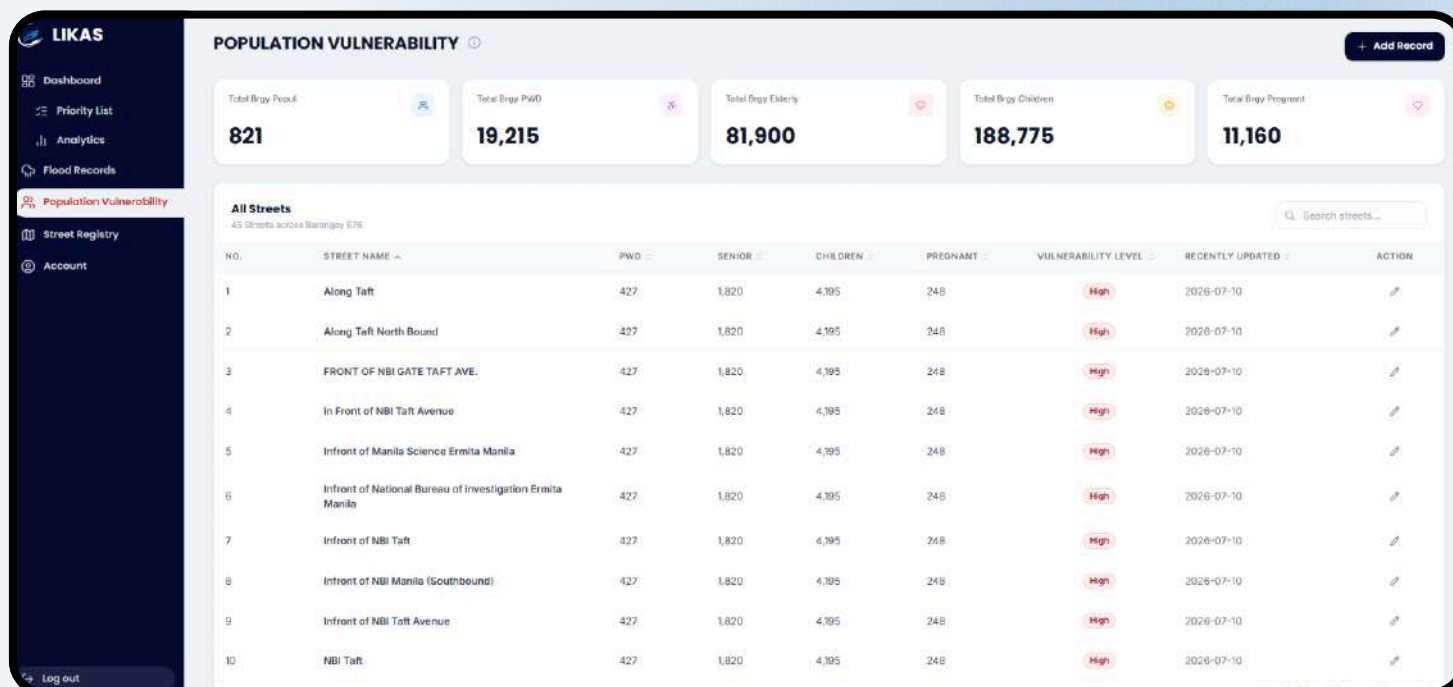
The Log Incident feature allows Barangay users to record and manage flood incidents within their assigned barangay.

1. Click Log Incident from the Flood Records page.
2. Pin the exact location of the flood incident on the interactive map.
3. Ensure that the selected location is within the barangay boundary. If the pin is placed outside the boundary, the system will display an "Outside barangay boundary" warning and prevent you from proceeding.
4. Once a valid location is selected, click Next.
5. Enter the required incident details, including:
 - Date
 - Time
 - Street
 - Flood Depth
 - Cause of Flooding

Remarks (optional image attachment)

6. Click Review Details to verify the entered information.
7. Review the incident summary, then click Confirm to save the new flood incident.
8. After the incident is successfully recorded, verify that it appears in the Flood Records list with the correct information.

Step-by-Step Usage – BARANGAY



POPULATION VULNERABILITY

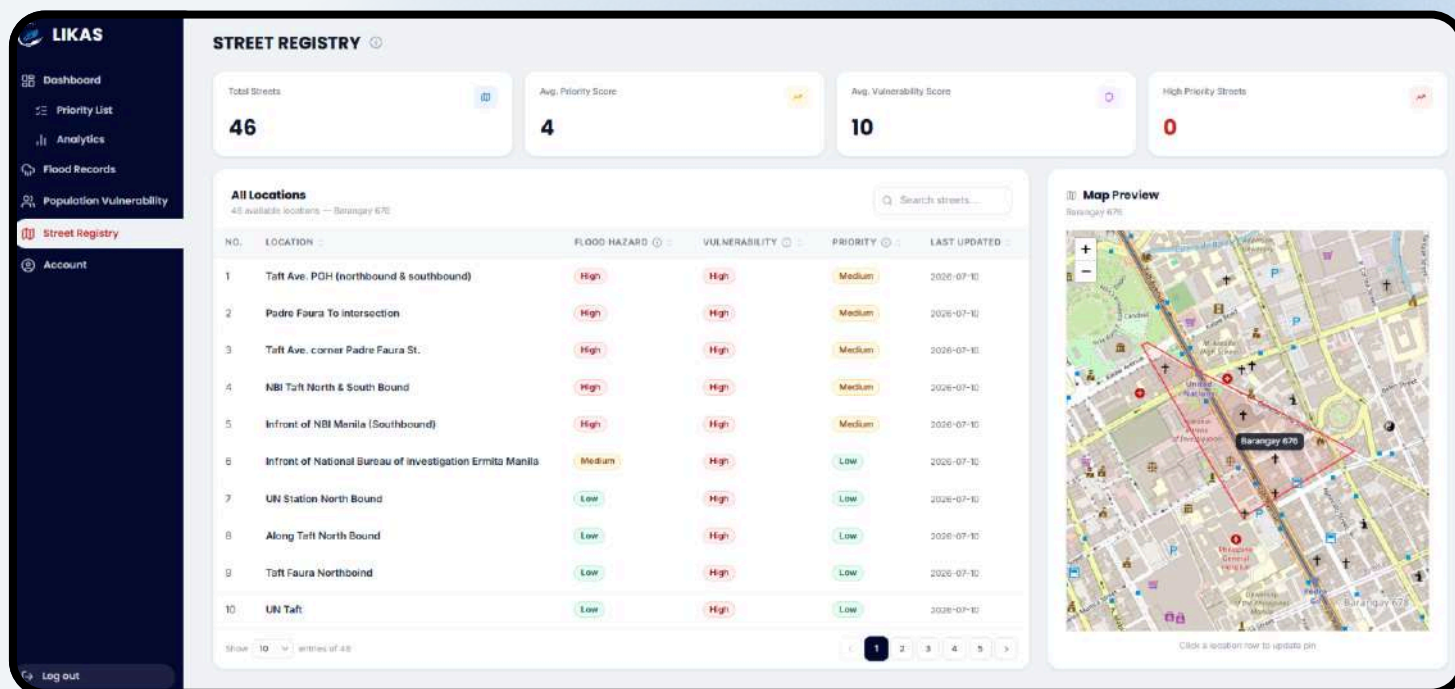
The Population Vulnerability page displays demographic information for each street within the assigned barangay, including vulnerable population groups used in the system's priority assessment.

1. View the summary cards to monitor the total barangay population, persons with disabilities (PWDs), senior citizens, children, and pregnant women.
2. Review the street list to view the population counts for each vulnerable group.
3. Use the Search bar to quickly locate a specific street.
4. Check the Vulnerability Level assigned to each street based on its demographic data.
5. Click the Edit icon under the Action column to update the street's population vulnerability information.
6. Click Add Record to create a new population vulnerability record for a street.

NOTE !

Changes made to a street's vulnerability data are automatically reflected in the system's priority assessment and ranking process.

Step-by-Step Usage – BARANGAY



STREET REGISTRY

The Street Registry page allows Barangay users to view and monitor all registered streets within their assigned barangay, including their flood hazard, vulnerability, and priority classifications.

1. View the summary cards to monitor the total number of streets, average priority score, average vulnerability score, and high-priority streets.
2. Use the Area filter to display specific street locations.
3. Browse the list of registered streets and review their Flood Hazard, Vulnerability, and Priority levels.
4. Use the Search bar to quickly locate a specific street.
5. Click a street from the list to highlight and preview its location on the map.

NOTE !

Street classifications are automatically updated based on flood incidents and population vulnerability data.

Step-by-Step Usage – BARANGAY

LIKAS

- Dashboard
- Priority List
- Analytics
- Flood Records
- Population Vulnerability
- Street Registry**
- Account

STREET REGISTRY

Total Streets: **46** | Avg. Priority Score: **4** | Avg. Vulnerability Score: **10** | High Priority Streets: **0**

All Locations
46 available locations — Barangay 676

NO.	LOCATION	FLOOD HAZARD	VULNERABILITY	PRIORITY	LAST UPDATED
1	Taft Ave. PGH (northbound & southbound)	High	High	Medium	2026-07-10
2	Padre Faura To Intersection	High	High	Medium	2026-07-10
3	Taft Ave. corner Padre Faura St.	High	High	Medium	2026-07-10
4	NBI Taft North & South Bound	High	High	Medium	2026-07-10
5	Infront of NBI Manila (Southbound)	High	High	Medium	2026-07-10
6	Infront of National Bureau of Investigation Ermita Manila	Medium	High	Low	2026-07-10
7	UN Station North Bound	Low	High	Low	2026-07-10
8	Along Taft North Bound	Low	High	Low	2026-07-10
9	Taft Faura Northbound	Low	High	Low	2026-07-10
10	UN Taft	Low	High	Low	2026-07-10

Show: 10 | 1 of 46

Map Preview
Barangay 676

Click a location row to update pin

LIKAS

- Dashboard
- Priority List
- Analytics
- Flood Records
- Population Vulnerability
- Street Registry**
- Account

46 | **4** | **10** | **0**

Showing: Barangay 676 — NBI Taft North & South Bound Medium

All Locations
46 available locations — Barangay 676

NO.	LOCATION	FLOOD HAZARD	VULNERABILITY	PRIORITY	LAST UPDATED
1	Taft Ave. PGH (northbound & southbound)	High	High	Medium	2026-07-10
2	Padre Faura To Intersection	High	High	Medium	2026-07-10
3	Taft Ave. corner Padre Faura St.	High	High	Medium	2026-07-10
4	NBI Taft North & South Bound	High	High	Medium	2026-07-10
5	Infront of NBI Manila (Southbound)	High	High	Medium	2026-07-10
6	Infront of National Bureau of Investigation Ermita Manila	Medium	High	Low	2026-07-10
7	UN Station North Bound	Low	High	Low	2026-07-10
8	Along Taft North Bound	Low	High	Low	2026-07-10
9	Taft Faura Northbound	Low	High	Low	2026-07-10
10	UN Taft	Low	High	Low	2026-07-10

Show: 10 | 1 of 46

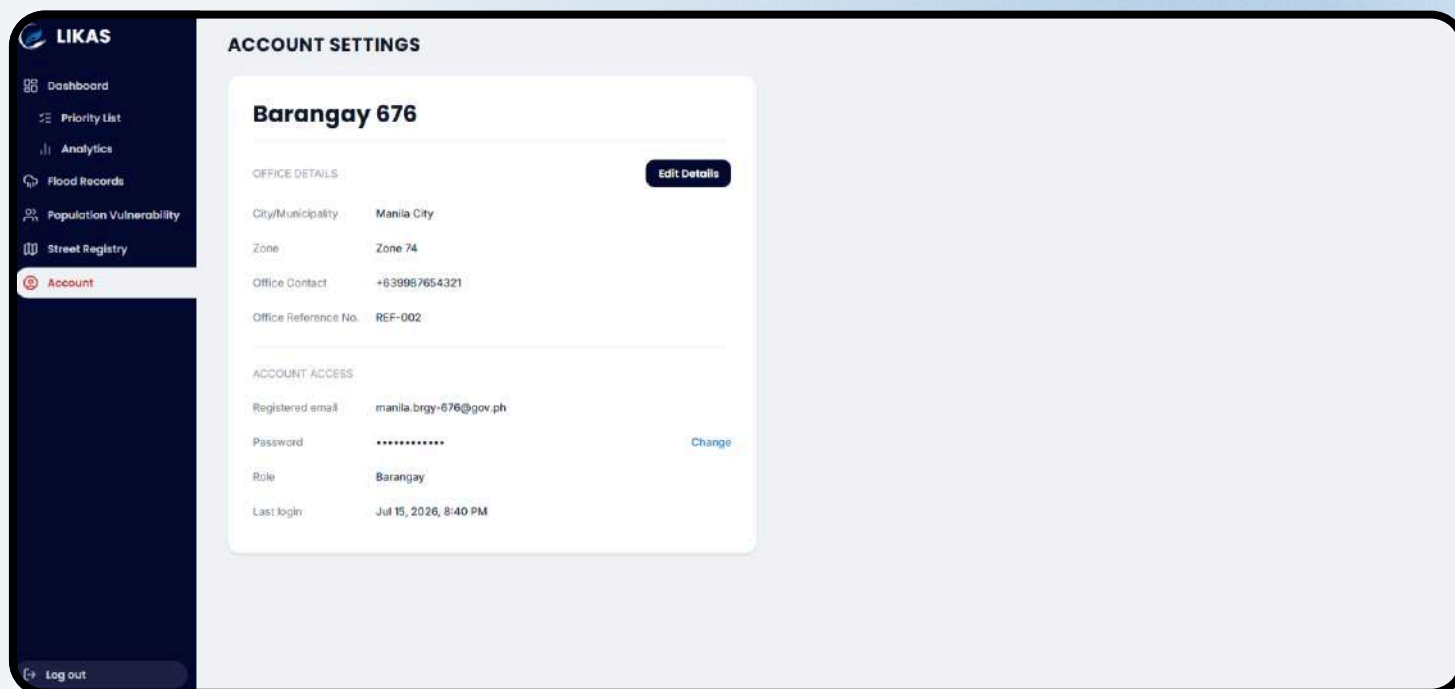
Map Preview
Barangay 676 — NBI Taft North & South Bound

Click a location row to update pin

NOTE !

Street classifications are automatically updated based on flood incidents and population vulnerability data.

Step-by-Step Usage – BARANGAY



ACCOUNT SETTINGS

The Account Settings page allows Barangay users to view and manage office information and account credentials used in the system.

1. Review the Office Details section, including the City/Municipality, Zone, Office Contact Number, and Office Reference Number.
2. Click Edit Details to update the office information displayed in the system.
3. View the Account Access section to check the registered email address, password hash, role, and last login information.
4. Click Change beside the password field to update your account password.

NOTE !

Only authorized account holders can modify account details and credentials. Changes made to office information are reflected throughout the system where applicable.



LIKAS

ACCESSIBILITY FEATURES

INTERACTIVE MAPS

The Interactive Maps feature allows users to visualize barangay boundaries and registered street locations within Manila. The map supports zooming and panning for easier navigation and automatically displays the selected street location when a street is chosen from the Street Registry page. This helps users quickly identify flood-prone and prioritized locations within the mapped barangay area.



659 — Antonio Villegas Street High

Barangays

FLOOD HAZARD

VULNERABILITY

PRIORITY

LAST UPDATED

Sta Cruz Manila	High	High	High	2026-07-10
Street	High	Medium	High	2026-07-10
Padre Fauro St.	High	High	Medium	2026-07-10
South Bound	High	High	Medium	2026-07-10
Intersection	High	High	Medium	2026-07-10
Manila (Southbound)	High	High	Medium	2026-07-10
Northbound & southbound)	High	High	Medium	2026-07-10
Street Brgy 598 Damka Manila	High	High	Medium	2026-07-10
Brgy 598 Damka Sta. Mesa	High	High	Medium	2026-07-10
Co	High	High	Medium	2026-07-10

1

2

3

4

5

>

Map Preview

Barangay 659 — Antonio Villegas Street

Barangay 659 — Antonio Villegas Street

A map preview showing the location of Barangay 659 — Antonio Villegas Street. A red pin is placed on the map, and a red bounding box highlights the area around it. The map includes labels for various landmarks and streets.

Click a location row to update

COLOR CODED LEVELS

	PRIORITY LEVEL	ACTION
St. Cruz Manila	High	View Map
Intersection	High	View Map
South Bound	Medium	View Map
Padre Faura St.	Medium	View Map
Northbound & southbound	Medium	View Map
Southbound	Medium	View Map
East Brgy 598 Damka Manila	Medium	View Map
West Brgy 598 Damka Sta. Mesa	Medium	View Map
	Medium	View Map

The system uses color-coded labels to visually represent flood hazard, vulnerability, and priority levels. High-risk locations are highlighted in red, medium-risk locations in yellow, and low-risk locations in green. These visual indicators help users quickly identify and compare areas that may require greater attention, monitoring, or resource allocation. Color-coded levels are automatically updated based on the latest flood records and vulnerability assessment data.

Barangay 659 — Antonio Villegas Street High

	FLOOD HAZARD	VULNERABILITY	PRIORITY	LAST UPDATED
St. Cruz Manila	High	High	High	2026-07-10
Street	High	Medium	High	2026-07-10
Padre Faura St.	High	High	Medium	2026-07-10
South Bound	High	High	Medium	2026-07-10
Intersection	High	High	Medium	2026-07-10
Manila (Southbound)	High	High	Medium	2026-07-10
Northbound & southbound	High	High	Medium	2026-07-10
East Brgy 598 Damka Manila	High	High	Medium	2026-07-10
West Brgy 598 Damka Sta. Mesa	High	High	Medium	2026-07-10
Co	High	High	Medium	2026-07-10

Map Preview

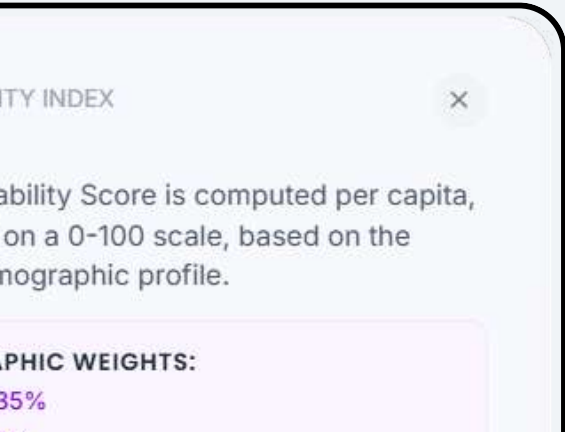
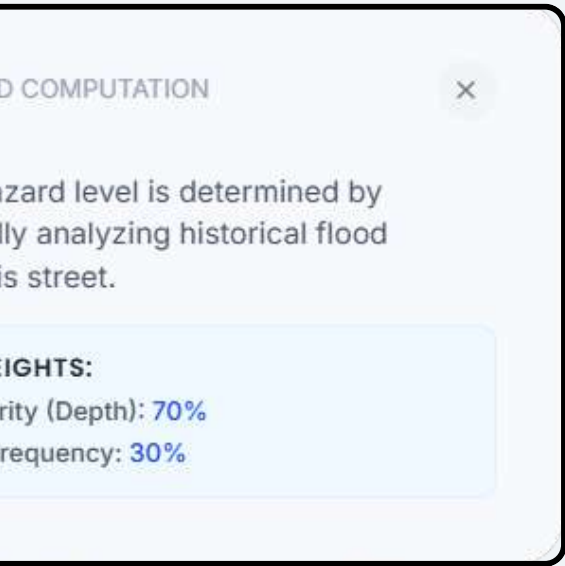
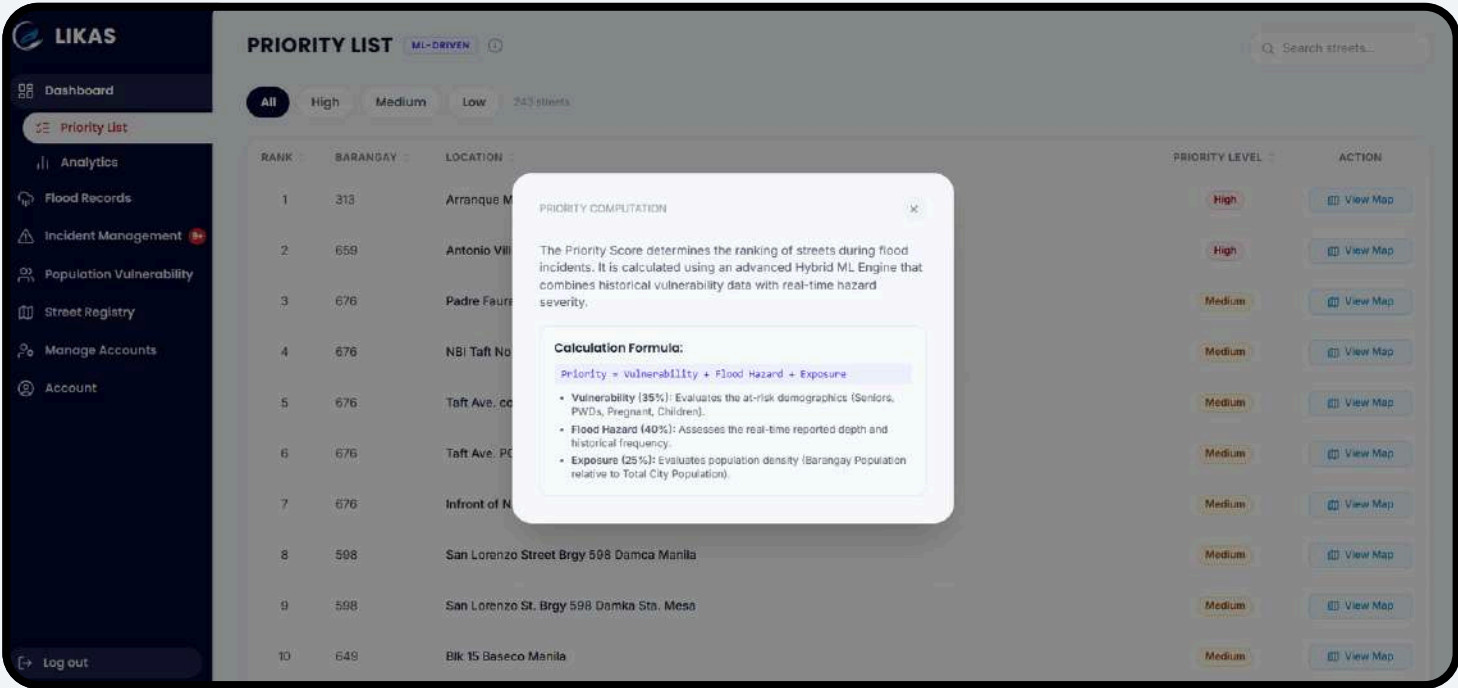
Barangay 659 — Antonio Villegas Street

Barangay 659 — Antonio Villegas Street

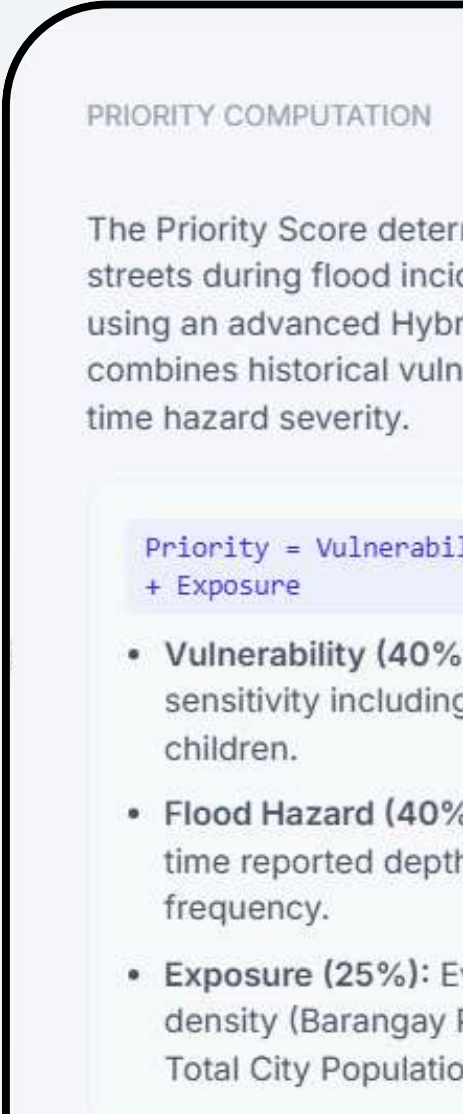


Click a location row to update

INFORMATION TOOLTIPS



Information tooltips provide additional explanations about how flood hazard levels, vulnerability scores, and priority rankings are calculated within the system. Users can click the information icons to view the factors, formulas, and data used in each assessment. These tooltips help users better understand the system's evaluation process and promote transparency in the generated results.





LIKAS

EXPECTED BUGS & RESOLUTIONS

EXISTING BUGS & RESOLUTIONS

The following guide details common operational errors and their designated troubleshooting protocols. Administrators and users are advised to follow these recommended resolution steps before escalating persistent issues to the primary technical support team.

Network Error

The map fails to load or real-time data stops syncing.

Check the local internet connection. If the connection is stable, refresh the application. Ensure the backend server's CORS settings are properly configured to accept requests from the frontend.

Authentication Error

Users are unexpectedly logged out or cannot log in.

This usually indicates an expired JSON Web Token (JWT). Simply return to the login page and re-authenticate.

EXISTING BUGS & RESOLUTIONS

Inaccurate Data Recognition

User-submitted reports show impossible flood depths.

Administrators can manually flag or override anomalous data in the "Report Management" tab to prevent the algorithm from prioritizing false alarms.

Geolocation & Pin Accuracy Issues

The map preview drops the street pin in the wrong location

Ensure the browser has been granted Location Tracking permissions. If a submitted street is pinning in the wrong city, verify that the user entered the correct exact street name so the Geocoding API can properly translate the address into precise Latitude/Longitude coordinates

Report Export / PDF Generation Failure

Clicking "Download PDF Report" on the Analytics page does nothing, or generates a corrupted file.

This is typically caused by the charts not fully rendering before the screen capture initiates. Ensure you wait at least 1-2 seconds after changing a date filter before clicking download. If the issue persists, clear your browser cache.

Technical Support

For questions, concerns, or assistance regarding the LIKAS Prioritization System, please contact the system administrator or the designated DRRMO office.

Office: Manila City DRRMO

Email: manila.mdrmo@gov.ph

Contact Number: +639182781728

You may also contact the system developers for technical issues, bug reports, system maintenance concerns, or assistance related to system functionality and troubleshooting.

Email:

patriciabalansag01@gmail.com,
krishaannname.escano@gmail.com,
stephaniealveroligon@gmail.com
acfsuico@gmail.com

To ensure timely assistance, please provide complete details regarding the issue encountered, including the affected feature and any relevant screenshots or error messages.